

YUE2: UNIFYING SYMBOLIC AND AUDIO MUSIC GENERATION AT FRONTIER QUALITY*



GitHub: <https://github.com/multimodal-art-projection/YuE>
Hugging Face: <https://huggingface.co/m-a-p/YuE2-3B>
Demo Page: <https://map-yue2.github.io/>

ABSTRACT

Symbolic models make melody, harmony, rhythm, and form explicit but typically stop before a finished recording; audio models produce complete songs while leaving composition implicit. We introduce **YuE2**, which unifies symbolic and audio music generation at **frontier quality** through **symbolic planning**. A single AR-NAR Mixture-of-Transformers (MoT) [14] first writes a readable score specifying melody and harmony, expands it into semantic music tokens, and realizes it as full-song audio. In comparisons using the same checkpoint, experts prefer symbolic planning for overall quality and musicality, with 49.3% of overall preferences versus 34.6% without planning. Experts also favor the unified model over a separate language model and diffusion Transformer [81]. On WildSongBench, YuE2 scores 6.73 on SongBench [77] Global Avg, exceeding all evaluated public baselines. Selecting from eight candidates (best-of-8), YuE2 reaches 6.96, the highest observed mean among all evaluated systems. Expert listening further establishes its competitiveness with proprietary song generators, favoring best-of-8 over Suno v4.5 [68] and yielding nearly balanced preferences against Suno v5 [65]. To learn this generation process from recordings without aligned scores, we introduce **MERT2** and **SheetSage2** to supply semantic and symbolic supervision. MERT2 sets a new state of the art in music representation learning, surpassing previous best results on 14 of 15 MARBLE [87] metrics; SheetSage2 leads 12 of 15 benchmark-metric pairs in our lead-sheet transcription comparison. The same checkpoint follows score edits while largely preserving unedited musical content and generates **zero-shot covers** without cover-specific training. Its readable score also enables **agentic music editing**, with external language models translating user feedback into revisions of the composition.

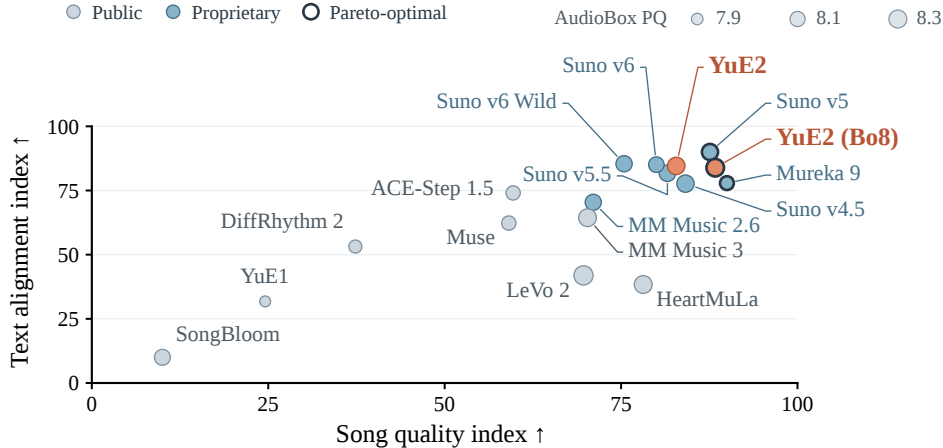


Figure 1: **YuE2 achieves frontier song quality, competitive with the evaluated proprietary systems.** Automatic evaluation on WSB (192 prompts), including Suno v6 and v6 Wild [66]. The quality index combines standardized SongBench [77] and SongEval [86] Global Avg; alignment combines standardized MuQ-MuLan [91], AllMusicCaps [3], and our Qwen3-Omni [80] score. Bubble area: AudioBox production quality (PQ) [70]; black outlines: the observed Pareto front on these two indices. Both YuE2 settings use symbolic planning; Bo8 means best-of-8. Appendix A.4 gives index construction and weight sensitivity; Figure 7 reports expert preferences.

*Full author list and affiliations on page 22.

1 INTRODUCTION

A finished song is a dense acoustic object, but creating it need not be one dense decision. Music has long been described through representations that range from textual knowledge and symbolic notation to performance information and audio signals [12, 73]. Figure 2 organizes these levels by the decisions they settle. Text and lyrics state intent. A score commits to melody, harmony, rhythm, and form. Performance and audio representations then resolve timing, articulation, timbre, expression, and production. We call generation through these levels *progressive musical commitment*: each stage settles additional decisions while leaving the remaining choices to the stages below. The space between a score and its sound is room for interpretation.

Seen through this hierarchy, many generators expose only one side. Symbolic models make composition readable but stop before a finished recording [29, 8]. Audio models produce the recording but usually skip an explicit score, leaving composition implicit [1, 54]. Can a single model traverse this hierarchy, making composition readable and editable while improving the quality of the finished song?

We introduce YuE2, a single model that composes in symbols and performs in audio. Writing the composition first improves perceived song quality. The model first writes a composition containing melody, chords, key, meter, tempo, and form—a step we call *symbolic composition planning*. It then supplies detail left unspecified by the score [79] through semantic music tokens and continuous acoustic latents for stereo decoding. One AR–NAR Mixture-of-Transformers [14] predicts the symbolic and semantic sequences causally and realizes the acoustic sequence through bidirectional flow matching [47]. With symbolic planning, YuE2 also reaches frontier full-song quality, competitive with the evaluated proprietary systems.

Ordinary audio corpora lack the aligned notes, chords, beats, and sections needed to learn this hierarchy [22, 8]. We introduce MERT2 and SheetSage2 to construct symbolic and semantic supervision from recordings. SheetSage2 recovers readable lead sheets, while the MERT2 tokenizer supplies semantic music tokens.

We test the contribution of symbolic planning by comparing generation with and without planning using the same checkpoint, prompts, sampling budget, and decoder. Experts prefer planned generation for overall quality and musicality. For overall quality, 49.3% of judged responses favor planning and 34.6% favor generation without planning; the rest are ties. With melody-and-chord planning in both systems, they also prefer YuE2’s unified MoT to a separate language model and diffusion Transformer (LM+DiT) [81] for overall quality, musicality, audio quality, vocals, and accompaniment.

In expert listening, YuE2 (best-of-8) is preferred over Suno v4.5 [68] and receives nearly balanced overall preferences against Suno v5 [65] and Mureka 9 [51]. Audio quality is a particular strength: its mean tie-adjusted preference across six proprietary systems is 58.9%. On WildSongBench (WSB), YuE2 leads the evaluated public systems on SongBench [77] Global Avg (6.7316), while best-of-8 achieves the highest observed mean among all evaluated systems (6.9632; Section 4.1).

The same YuE2 checkpoint renders its own compositions, revised scores, and transcriptions of existing recordings. Score–audio comparisons show strong agreement between the planned and rendered melody and harmony (Section 5.1). Controlled editing experiments show that score edits change the targeted melody and harmony while largely preserving unedited musical content (Section 5.2). Without cover-specific training, YuE2 with full scores surpasses both evaluated cover systems on all eight work-identity retrieval measures, AudioBox [70] production quality, and SongBench Musicality across 948 unseen works (Section 5.3). The readable score also enables *agentic music editing*: an external language-model agent turns user feedback into explicit revisions of the composition, which YuE2 renders as a new recording. We demonstrate this interaction in a case study (Section 5.4).

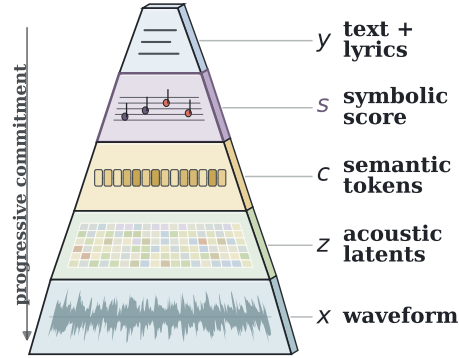


Figure 2: YuE2 progressively commits to musical detail while keeping the symbolic score explicit and inspectable.

Contributions.

1. We introduce YuE2, which unifies symbolic and audio music generation through symbolic planning and reaches frontier full-song quality. Its readable score enables controlled editing, zero-shot cover generation, and agentic music editing with the same checkpoint.
2. We show that symbolic planning improves perceived song quality and that, with melody-and-chord planning, experts prefer a unified MoT to separate LM+DiT.
3. We introduce MERT2 and SheetSage2 to supply semantic and symbolic supervision from recordings. MERT2 establishes a new state of the art on MARBLE [87], surpassing previous best results on 14 of 15 metrics across nine tasks. SheetSage2-AR leads 12 of 15 benchmark–metric pairs in our unified full-song transcription comparison. Trained entirely on labels from SheetSage2-Prober, it also surpasses its label generator on 10 of 15 benchmark–metric pairs under the same evaluation protocol.

2 YUE2: COMPOSING IN SYMBOLS, PERFORMING IN AUDIO

2.1 GENERATION OVERVIEW

YuE2 represents composition explicitly and progressively realizes it through semantic tokens and acoustic latents. Let y denote text conditions and lyrics, and let

$$s = \text{symbolic composition}, \quad c = \text{semantic tokens}, \quad z = \text{continuous acoustic latents}. \quad (1)$$

The model factorizes generation as

$$p_{\theta}(s, c, z | y) = p_{\theta, \text{AR}}(s, c | y) p_{\theta, \text{NAR}}(z | y, s, c). \quad (2)$$

The sparse sequence s carries decisions a musician can read and change. The 25-Hz sequence c supplies a dense musical trajectory, including information that is awkward to write in a lead sheet. The 25-Hz latent sequence $z \in \mathbb{R}^{T \times 64}$ retains the acoustic detail needed by a separately trained 48-kHz stereo decoder. Training targets for all three representations are constructed offline from recordings (Section 3).

2.2 SYMBOLIC COMPOSITION AS A GENERATIVE STATE

The symbolic composition uses ABC, a text-based music notation, serialized with byte-pair encoding (BPE) [58]. Its headers specify tempo, meter, and key; barlines and section labels organize time and form; chord symbols specify harmony; and vocal and instrumental voices specify melody. When generating both a score and semantic tokens, the autoregressive stream produces

$$[\text{ABC_START}], s, [\text{ABC_END}], [\text{MUSIC_START}], c, [\text{MUSIC_END}] \quad (3)$$

after the text and lyric condition. The acoustic stream conditions on both the completed symbolic and semantic sequences.

ABC shares the ordinary text tokenizer, while semantic codes occupy a disjoint range. Type-specific output masks prevent the two spans from emitting each other’s symbols. ABC is sampled without grammar constraints.

2.3 AR–NAR MIXTURE-OF-TRANSFORMERS

Composition and acoustics require different information flow. The symbolic and semantic tokens are ordered discrete sequences and are predicted causally. Once those sequences are complete, every acoustic frame can use the full plan and bidirectional acoustic context. YuE2 implements both computations in one 28-layer backbone inspired by Mixture-of-Transformers [14] (architecture evaluation in Section 4.3). Each layer has separate AR and NAR normalization, query/key/value and output projections, and multilayer-perceptron (MLP) experts. The streams share the positional scheme and participate in one attention computation.

Let A and N denote discrete and acoustic positions. The hybrid mask is

$$\begin{array}{c|cc} \text{query} \backslash \text{key} & A & N \\ \hline A & \text{causal} & \text{blocked} \\ N & \text{full} & \text{bidirectional} \end{array}. \quad (4)$$

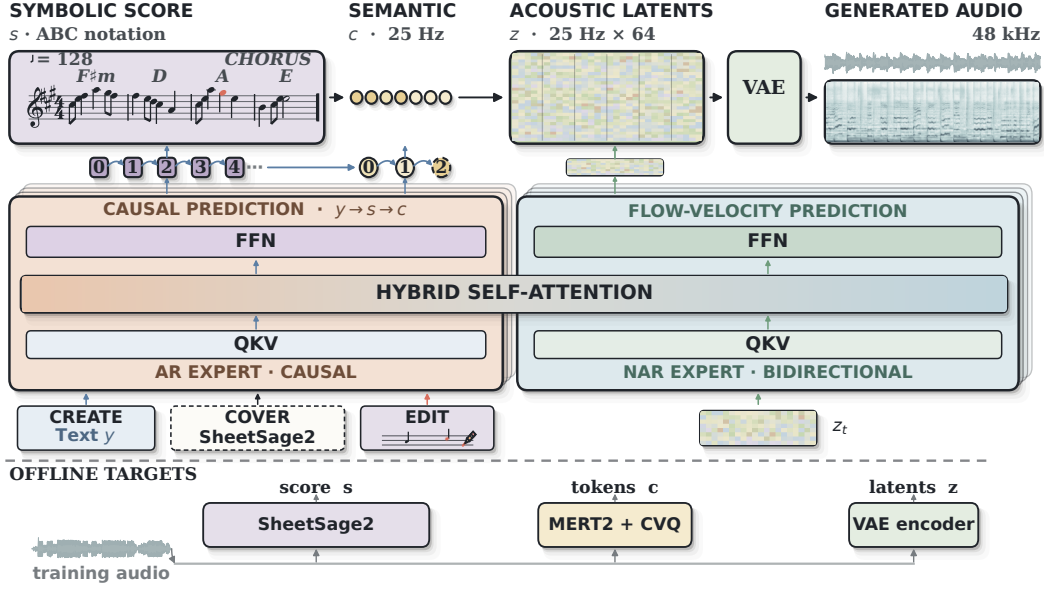


Figure 3: **Composing in symbols, performing in audio.** YuE2 generates an editable symbolic composition s , 25-Hz MERT2 semantic tokens c , and 25-Hz continuous acoustic latents z before 48-kHz stereo variational-autoencoder (VAE) decoding [38]. **Creation:** YuE2 generates s from the conditioning text. **Covering:** SheetSage2 extracts s from a reference recording. **Editing:** the user supplies a modified s . The same YuE2 checkpoint then generates the semantic and acoustic representations in all three modes. Within each of 28 layers, side-by-side AR and NAR experts use separate normalization, projections, and MLPs around one shared attention computation. The AR stream conditions on style and lyric text y and performs causal next-token prediction over s and c ; the NAR stream predicts flow velocity over z_t , where $t \in [0, 1]$ is flow time, with bidirectional latent attention.

Discrete predictions cannot read the acoustic target, while acoustic states attend to all text, score, and semantic tokens and communicate across the full latent sequence.

At an acoustic position, a projection of the noisy latent replaces the token embedding and is combined with time and frame-position embeddings. A linear head predicts the flow velocity. All frames are updated in parallel within each velocity evaluation.

2.4 TRAINING

The AR stream is trained with next-token cross-entropy over the symbolic and semantic spans,

$$\mathcal{L}_{\text{AR}} = -\frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \log p_{\theta}(w_i | w_{<i}), \quad (5)$$

where $\mathcal{A}$ contains generated payload tokens and their closing markers. For acoustic learning, let $t \in [0, 1]$ denote flow time, let z_0 be the clean latent, $\epsilon \sim \mathcal{N}(0, I)$, and $z_t = (1 - t)z_0 + t\epsilon$. Conditional flow matching [47] minimizes

$$\mathcal{L}_{\text{FM}} = \frac{1}{64|\mathcal{Z}|} \sum_{i \in \mathcal{Z}} \|v_{\theta}(z_t, t, y, s, c)_i - (\epsilon_i - z_{0,i})\|_2^2. \quad (6)$$

Here, $\mathcal{Z}$ indexes the non-padding acoustic target positions. The joint objective is

$$\mathcal{L} = 0.25\mathcal{L}_{\text{AR}} + \mathcal{L}_{\text{FM}}. \quad (7)$$

During training, we sample from four tasks (Appendix Table 14). The main setting includes the symbolic score, semantic tokens, and acoustic latents. The other three omit the score, the semantic tokens, or both. Training on all four settings allows the same checkpoint to generate with different available inputs and supports a matched comparison with and without symbolic planning.

The model is trained on approximately 346,000 hours of music. We pack whole songs into a 24,576-position context without splitting songs across training examples. Text and lyric conditioning can

be dropped separately or together during training to support classifier-free guidance and generation with missing conditions. The main model has approximately 3.58B parameters. Appendix C gives its architecture and optimization settings.

2.5 CREATION, EDITING, AND COVER GENERATION

The same YuE2 checkpoint supports song creation, editing, and cover generation through a shared score interface. The three operations differ in how the symbolic score s is obtained. For **creation**, YuE2 generates the score from text and lyrics before rendering the song. For **editing**, a revised score $\hat{s}$ is supplied as a fixed prefix, and YuE2 regenerates the downstream semantic tokens and acoustic latents to produce a new full-song performance.

For **cover generation**, the external SheetSage2 transcriber recovers a score $\hat{s}$ from a reference recording. YuE2 renders this score under a new style description using the same fixed-prefix interface. The score-to-audio mapping learned from individual recordings thus supports cover generation without training on original–cover pairs.

3 FROM RECORDINGS TO MUSICAL SUPERVISION

Training a model to compose before it renders requires two targets absent from ordinary recordings: a readable composition and a compact semantic token sequence. For each training recording x , three frozen analysis paths construct aligned views,

$$x \xrightarrow{\text{SheetSage2}} s, \quad x \xrightarrow{\text{MERT2 tokenizer}} c, \quad x \xrightarrow{\text{VAE}} z. \quad (8)$$

These frozen target constructors operate offline. Lead-sheet transcription uses bidirectional full-song context. For semantic tokenization, we follow Qwen-Music [81] and adapt a separate MERT2 branch with causal self-attention to produce compact music tokens for autoregressive prediction.

3.1 MERT2: MULTI-VIEW TARGETS AND FOUNDATION PRETRAINING

Complementary views as supervision. Encoders trained with different data and objectives need not describe a song in the same way: each makes some musical relations explicit and leaves others implicit. Yet the Platonic Representation Hypothesis proposes that independently learned representations can converge toward shared structure in the world [30]. Related evidence from SPEAR shows that stronger target features can improve a downstream representation learner [85]. Together, these observations motivate a discrete target that must support complementary views of the same recording.

Multi-view target synthesis. Figure 4A turns this idea into *Multi-View Target Synthesis*, an offline procedure that constructs MERT2’s targets before its four-stage curriculum begins. Frozen MuQ [91] and Qwen2-Audio-Instruct [9] encoders first map each recording to time-aligned features. A learned fusion network combines the two feature sequences in one shared bottleneck, which a four-level residual vector quantizer discretizes. Separate decoders must then reconstruct both source representation spaces from the same quantized state. We cache the resulting four code streams as MERT2’s prediction targets. Each code is thus jointly constrained by two views rather than copied from either encoder.

Foundation pretraining. These cached targets begin the four-stage MERT2 curriculum.¹ In Stage 1, *Foundation Pretraining*, a new audio encoder predicts the four code streams at masked frames. It converts 24-kHz mono audio into 128-bin log-mel features, subsamples them to 25 Hz with a ConvNeXt frontend [48], and processes the sequence with a 24-layer, 1,024-dimensional Conformer [27].

¹MERT2-30s: <https://huggingface.co/m-a-p/MERT-v2-30s>;
MERT2-FS: <https://huggingface.co/m-a-p/MERT-v2-FullSong>.

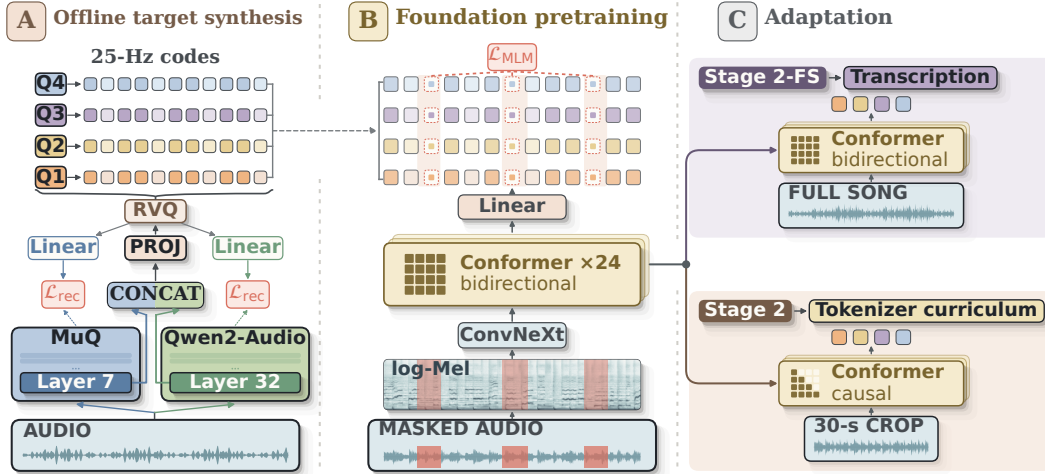


Figure 4: **MERT2 learns from discrete targets jointly derived from two encoder views of the same audio.** (A) **Multi-View Target Synthesis.** Frozen MuQ [91] and Qwen2-Audio-Instruct [9] encoders map the same recording to time-aligned features. A shared four-level residual vector quantizer produces four cached 25-Hz code streams. Separate decoders reconstruct both feature spaces from the same quantized representation. (B) **Stage 1 — Foundation Pretraining.** A bidirectional ConvNeXt–Conformer encoder [48, 27] is trained to predict these codes from masked audio. (C) **Adaptation.** Stage 2-FS — Full-Song Adaptation continues pretraining on complete recordings lasting 30–360 seconds while retaining bidirectional context, producing MERT2-FS for SheetSage2. Stage 2 — Causal Adaptation switches self-attention to a causal mask and continues along the semantic-tokenizer branch.

One foundation, two temporal regimes. After Foundation Pretraining, we adapt separate branches for full-song transcription and semantic tokenization. Stage 2-FS, *Full-Song Adaptation*, preserves bidirectional attention and extends the objective to complete recordings lasting 30–360 seconds; its MERT2-FS features support SheetSage2. The tokenizer branch instead enters Stage 2, *Causal Adaptation*, where self-attention is restricted to the current and earlier positions. It then proceeds through Stage 3, *Supervised Fine-Tuning*, and Stage 4, *Semantic Quantization*, to produce the deployed semantic tokenizer. Figure 4 shows the two branches adapted from the shared Stage 1 foundation for transcription and tokenization. Appendix Section B gives their objectives and layer choices; Section 6 evaluates the resulting representations separately.

3.2 SHEETSAGE2: FULL-SONG TRANSCRIPTION

SheetSage2 denotes the autoregressive transcriber SheetSage2-AR². It recovers an editable lead sheet from a complete recording, including melody, chords, key, beat, downbeat, meter, and section structure. The model uses a full-context MERT2-FS encoder with a frozen backbone and trainable low-rank adapters, followed by a six-layer autoregressive RoFormer decoder [64]. All attributes are generated in one chronological event sequence, anchored by beat timestamps quantized to 10 ms (Figure 5). A deterministic builder converts the events into ABC notation with explicit measures, rests, ties, and separate vocal and instrumental melody voices.

Generating aligned annotations. Training SheetSage2-AR requires a shared vocabulary and timing convention for multiple musical attributes. Existing music information retrieval (MIR) datasets provide complementary annotations with differing coverage and conventions. We first train the non-autoregressive *SheetSage2-Prober*, fine-tuning MERT2 through low-rank adapters and task-specific prediction heads. Its supervision mixes human-annotated datasets [16, 75, 52] and MIDI-rendered audio [35, 17]. We compute losses only for annotated attributes, enabling training on partially annotated recordings. Task-specific conditional random field (CRF)-based structured decoding [56] combines these neural scores with temporal constraints to recover discrete labels: rhythm decoding first establishes beats, downbeats, and meter, then key, chord, structure, and melody are decoded on the resulting grid. We apply the prober to real audio recordings to generate all SheetSage2-AR

²SheetSage2-AR checkpoint and inference code: <https://huggingface.co/m-a-p/SheetSage2>.

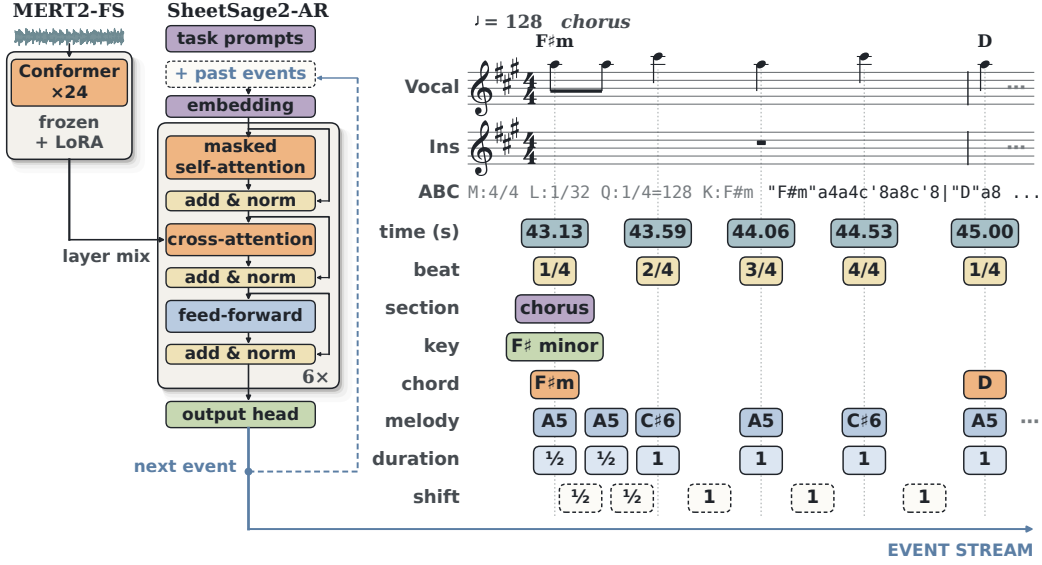


Figure 5: **SheetSage2-AR architecture and output representation.** A full-context MERT2-FS encoder with a frozen backbone and trainable low-rank adapters supplies a learned layer mixture to a six-layer autoregressive decoder. Task prompts select the target attributes, and grammar-constrained decoding produces a chronological sequence of timing, metrical, structural, harmonic, and melodic events. A deterministic builder converts the event sequence to ABC notation, which is rendered as a lead sheet. The right panel uses a reference bar to illustrate this representation. The excerpt is serialized independently, with the active key emitted at its first beat and omitted thereafter unless it changes.

training labels under a shared attribute vocabulary and timing convention. SheetSage2 supplies the symbolic training targets and reference compositions for YuE2.

3.3 MERT2 TOKENIZER: COMPACT SEMANTIC TARGETS

Following Qwen-Music [81], we adapt MERT2 with causal self-attention before supervised fine-tuning and quantization. This aligns the encoder’s attention pattern with the downstream generator’s left-to-right prediction order. The adaptation continues masked prediction of the same multi-view targets from the pretrained checkpoint. We then add lyric and mel/chroma supervision on full songs before introducing the discrete bottleneck used to produce YuE2’s semantic tokens. Figure 6 summarizes the training curriculum and deployed tokenizer.

Causal-attention adaptation. *Causal Adaptation* initializes from *Foundation Pretraining* and replaces bidirectional self-attention with a causal mask while retaining the four synthesized target streams and the masked-prediction objective. Tokenization runs offline, combining causal self-attention with noncausal convolution and temporal normalization.

Grounding before compression. The attention mask controls contextual aggregation; the supervised objectives specify which musical attributes the tokens should preserve. Stage 3, *Supervised Fine-Tuning*, trains the adapted encoder on complete recordings of up to 360 seconds with three complementary objectives. Connectionist temporal classification aligns the states with lyrics [25]; mel and chroma reconstruction train the states to predict spectral and pitch-class features. Their heads remain attached in Stage 4, so the same criteria continue to act after the representation becomes discrete.

A supervised bottleneck. Stage 4, *Semantic Quantization*, turns each 1,024-dimensional continuous state after Conformer layer 13 into one categorical decision. A learned input projection maps each state to 32 dimensions, and cosine-similarity assignment selects its token ID k_t from a 32,768-entry codebook. To keep rarely selected entries available for learning, we adapt the online clustered codebook of CVQ-VAE [90]. Usage-dependent updates move these entries toward current encoder features, while frequently selected entries learn through the VQ objective. Appendix B.1 gives the

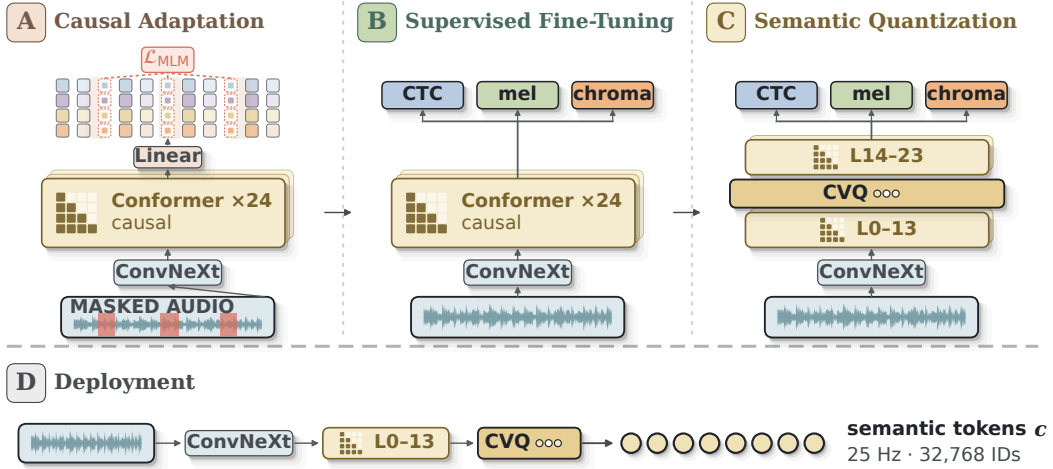


Figure 6: **MERT2 tokenizer training and deployment.** (A) **Causal Adaptation** initializes from Stage 1, masks temporal spans of the input audio, switches the 24-layer Conformer [27] to causal attention, and continues the Stage 1 objective: predicting all four cached target streams at masked positions under $\mathcal{L}_{MLM}$. (B) **Supervised Fine-Tuning** continues the adapted encoder on full songs and adds lyric connectionist temporal classification (CTC) [25] plus mel and chroma reconstruction. (C) **Semantic Quantization** inserts a learned 32-dimensional, 32,768-entry clustered vector quantizer (CVQ) between layers 13 and 14. Adapted from CVQ-VAE [90], its online feature-anchor update returns rarely selected entries to the assignment distribution. During Stage 4, layers 14–23 and the CTC, mel, and chroma heads remain above the quantizer, allowing their losses to supervise the quantized states. (D) **Deployment** retains the log-mel/ConvNeXt frontend [48], Conformer layers 0–13 with causal self-attention, and the quantizer, producing one semantic ID c every 40 ms (25 Hz; 375 bit/s) for autoregressive modeling and acoustic conditioning in YuE2.

update rule and loss settings. Codebook utilization exceeds 99% on the validation set. The selected embedding is projected back to the model width and passed through layers 14–23 and the lyric, mel, and chroma heads, supervising the discrete bottleneck with the Stage 3 tasks.

Tokenizer output and deployment. Training uses the complete network to shape the bottleneck; deployment keeps only the log-mel and ConvNeXt frontend, Conformer layers 0–13 with causal self-attention, and the quantizer. The resulting ID k_t is emitted every 40 ms. One 32,768-way decision carries 15 bits, giving a single 25-Hz semantic stream at 375 bits/s for YuE2’s autoregressive prediction. A separate VAE supplies continuous acoustic latents for waveform reconstruction. The semantic and acoustic streams share a 25-Hz clock and the same recording boundary.

3.4 CONSTRUCTING ALIGNED TRAINING SEQUENCES

To obtain the continuous acoustic targets z , we separately train an Oobleck variational autoencoder (VAE) following Stable Audio Open [19]. Its fully convolutional encoder uses six residual down-sampling blocks to compress 48-kHz stereo audio by a factor of 1,920 into a 64-dimensional variational bottleneck at 25 Hz. A mirrored decoder reconstructs the waveform through transposed convolutions and residual blocks.³

Each example concatenates text and lyric conditions with the available symbolic and semantic targets and aligns the acoustic latent sequence at the recording boundary. Symbolic tokens share the text vocabulary; semantic IDs occupy a disjoint range; acoustic latents remain continuous. This sequence format lets one checkpoint learn the complete $s \rightarrow c \rightarrow z$ trajectory and matched ablations in which s , c , or both are omitted.

³Listening previews: <https://huggingface.co/m-a-p/YuE2-Vae>;
Benchmark evaluation: <https://huggingface.co/m-a-p/YuE2-Vae-legacy>.

4 FRONTIER SONG QUALITY WITH EXPLICIT COMPOSITION

YuE2 reaches frontier full-song quality while exposing composition as a generated score. We compare the complete system with public and proprietary generators, then test two design choices: whether composing a score improves the finished song, and whether jointly learning semantic-token prediction and acoustic generation improves quality over training separate models.

4.1 FRONTIER FULL-SONG QUALITY ON WILDSongBENCH

Evaluation protocol. We assess song quality, prompt adherence, and lyric accuracy using automatic evaluators. The system comparison uses WildSongBench (WSB)⁴, a 192-prompt evaluation of in-the-wild user requests spanning 15 genre buckets. Its source-language labels comprise 94 Chinese and 98 English prompts, with naturally code-switched and multilingual requests retained. The user-prompt text and lyrics are exposed to each model through its native interface. We report the Musicality dimension and global average from SongBench [77] and SongEval [86], Qwen3-Omni-based prompt alignment (Q3O) [80], and phoneme error rate (PER). SongEval Avg is the mean of Clarity, Coherence, Memorability, Musicality, and Naturalness. SongBench Avg is the mean of all seven SongBench dimensions: Musicality, Melody, Arrangement, Structure, Instrumental, Mixing, and Vocal. Q3O is the prompt-weighted 0–5 aggregate over the applicable vocal, genre, mood, instrumentation, tempo, and key constraints; Appendix A.1 gives its calculation.

We also report production quality (PQ) from AudioBox Aesthetics [70] and audio–text alignment from MuQ–MuLan [91] (MuLan in the tables) and AllMusicCaps [3]. These two embedding metrics use the same standardized style description for each prompt, without lyrics. All nine reported metrics cover all 192 selected songs per system; Appendix A.2 gives preprocessing and aggregation details.

The public comparison covers YuE1 [88], SongBloom [83], LeVo 2 [44], ACE-Step 1.5 [23], HeartMuLa [84], DiffRhythm 2 [36], Muse [32], and MiniMax Music 3 [50], using its publicly released weights and official caption rewriter. For every system, the standard comparison selects the lower-PER output from two candidates per prompt.⁵ YuE2, the publicly released baselines, and Suno v6 and v6 Wild [66] use the lowest PER from four automatic speech-recognition (ASR) transcriptions per candidate. YuE2 best-of-8 selects from eight candidates, prioritizing SongBench Musicality, then Q3O, then PER. Both reported YuE2 settings use symbolic planning: the model generates a melody-and-chord ABC score before the semantic and acoustic representations. SongBloom is the sole input exception: its public interface cannot consume a text style prompt and instead requires the same held-out 10-s prompt audio for every item.

System comparison. Tables 1 and 2 separate proprietary and public systems while retaining the same nine metrics. The proprietary comparison includes Suno v4.5 [68], Suno v5 [65], Suno v5.5 [62], Suno v6 and v6 Wild [66], MiniMax Music 2.6 [49], and Mureka 9 [51].

YuE2 leads the evaluated public systems on SongBench Musicality (5.9075) and Global Avg (6.7316) with the standard two-candidate budget. At 3.58B parameters, it uses fewer parameters than six of the eight public baselines (Appendix A.5, Figure 11). With eight candidates, YuE2 (best-of-8) achieves the highest observed means across all evaluated systems on SongBench Musicality (6.2666) and Global Avg (6.9632). AudioBox PQ, which is not used for candidate selection, provides complementary evidence: YuE2 and YuE2 (best-of-8) score 8.2598 and 8.2714, respectively, exceeding every evaluated proprietary system.

On SongEval, YuE2 (best-of-8) ranks second among the evaluated public systems on Musicality (4.2506) and Avg (4.2960), behind HeartMuLa (4.5329 and 4.5519, respectively). HeartMuLa reports using SongEval and AudioBox in post-training [84].

⁴Prompts, reference scores, and evaluation code: <https://huggingface.co/datasets/m-a-p/WildSongBench>.

⁵The two-candidate budget mirrors proprietary interfaces that return two songs per request. ASR can produce different transcriptions of the same recording; taking the lowest PER across repeated transcriptions is intended to reduce the influence of occasional recognition failures.

Table 1: **Comparison with proprietary systems on WildSongBench (WSB).** All metrics cover 192 prompts. Mus.: Musicality; Avg: mean of all SongBench [77] or SongEval [86] dimensions; PQ: AudioBox production quality [70]; MuLan: MuQ-MuLan [91]; AMCaps: AllMusicCaps [3]; Q3O: Qwen3-Omni prompt adherence [80]; PER: phoneme error rate. Both YuE2 settings use symbolic planning. The standard comparison selects the lower-PER output from two candidates per prompt for every system; YuE2 best-of-8 (Bo8) selects from eight, prioritizing SongBench Musicality, then Q3O, then PER. Appendix A.3 details selection. MM abbreviates MiniMax. Bold and underlining mark the best and second-best observed means; higher is better except PER.

Model	SongBench		SongEval		AudioBox	Text alignment			Lyrics
	Mus. $\uparrow$	Avg $\uparrow$	Mus. $\uparrow$	Avg $\uparrow$	PQ $\uparrow$	MuLan $\uparrow$	AMCaps $\uparrow$	Q3O $\uparrow$	PER $\downarrow$
Suno v5 [65]	5.9918	6.8721	4.3051	4.3579	8.1698	0.5428	0.4353	4.5907	0.0810
Suno v4.5 [68]	5.8317	6.6995	<u>4.3198</u>	<u>4.3666</u>	8.2541	0.5022	0.3873	4.4149	0.0580
Suno v5.5 [62]	5.8087	6.7150	4.1497	4.2152	8.1955	<u>0.5089</u>	0.3917	4.5914	<u>0.0596</u>
Suno v6 [66]	5.6558	6.5562	4.2635	4.3086	8.1296	0.4916	0.4305	4.6258	0.0758
Suno v6 Wild [66]	5.5644	6.4195	4.1716	4.2199	8.1785	0.4999	<u>0.4316</u>	4.5898	0.0745
MM Music 2.6 [49]	5.4437	6.3222	4.0560	4.0985	8.1711	0.4251	0.3670	4.5688	0.2455
Mureka 9 [51]	<u>6.0488</u>	<u>6.9377</u>	4.3619	4.4111	8.0226	0.4394	0.4102	4.6368	0.1169
YuE2	5.9075	6.7316	4.2151	4.2625	<u>8.2598</u>	0.5068	0.4054	<u>4.6819</u>	0.0844
YuE2 (Bo8)	6.2666	6.9632	4.2506	4.2960	8.2714	0.5051	0.3980	4.7009	0.0979

Table 2: **Comparison with publicly released systems on WSB.** Prompts, metrics, abbreviations, and YuE2 selection budgets follow Table 1. Both YuE2 rows use symbolic planning. HeartMuLa[†] reports using SongEval and AudioBox during post-training to filter supervised fine-tuning data and construct DPO preference pairs [84]. Bold and underlined values indicate the best and second-best scores across all rows, respectively. Appendix A.3 gives the baseline protocols.

Model	SongBench		SongEval		AudioBox	Text alignment			Lyrics
	Mus. $\uparrow$	Avg $\uparrow$	Mus. $\uparrow$	Avg $\uparrow$	PQ $\uparrow$	MuLan $\uparrow$	AMCaps $\uparrow$	Q3O $\uparrow$	PER $\downarrow$
YuE1 [88]	4.0847	4.9165	3.1524	3.2150	7.8683	0.2623	0.2882	3.7301	0.3638
SongBloom [83]	3.4493	4.2350	3.2048	3.2051	8.1539	0.2697	0.1926	3.0287	0.1919
LeVo 2 [44]	5.4590	6.3247	3.9819	4.0234	8.3966	0.3542	0.2680	3.9458	0.2612
ACE-Step 1.5 [23]	5.1588	6.0118	3.8051	3.8465	8.0518	0.4372	0.3869	4.5809	<u>0.0746</u>
HeartMuLa [†] [84]	5.4963	6.2483	4.5329	4.5519	<u>8.2933</u>	0.3823	0.2786	3.4907	0.1071
DiffRhythm 2 [36]	4.4775	5.2428	3.4711	3.5245	7.9782	0.3782	0.3255	4.0870	0.1841
Muse [32]	5.1692	6.0349	3.7362	3.7887	8.0517	0.3937	0.3466	4.4038	0.3342
MM Music 3 [50]	5.3482	6.2830	4.0431	4.0994	8.2825	0.3928	0.3609	4.4362	0.0627
YuE2	<u>5.9075</u>	<u>6.7316</u>	4.2151	4.2625	8.2598	0.5068	0.4054	<u>4.6819</u>	0.0844
YuE2 (Bo8)	6.2666	6.9632	<u>4.2506</u>	<u>4.2960</u>	8.2714	<u>0.5051</u>	<u>0.3980</u>	4.7009	0.0979

For text alignment, YuE2 leads the evaluated public systems on MuLan (0.5068) and AllMusicCaps (0.4054); Suno v5 leads the full comparison on both metrics. For lyric accuracy, YuE2 has a PER of 0.0844; the lowest public-system PER is 0.0627 for MiniMax Music 3, followed by 0.0746 for ACE-Step 1.5.

Listening evaluation protocol. The Arena export records 351 participations: 289 online sessions recruited through X, GitHub, and Hugging Face, and 62 paid annotators with conservatory training or audio AI research backgrounds. The analysis snapshot contains 4,439 retained pairwise evaluations across three experiments on 192 prompts. Experts assessed anonymized audio pairs for overall quality, musicality, text alignment, audio quality, vocals, and accompaniment. Overall quality reflects preference for the song as a whole; musicality concerns its composition and artistic appeal; audio quality concerns acoustic fidelity and mixing. Appendix D.4 defines all six criteria and details

recruitment, listening requirements, and consistency checks, with statistical methods and confidence intervals in Appendix D.4.4.

Expert listening. For overall quality, 57.3% of judged responses favor YuE2 (best-of-8) and 30.5% favor Suno v4.5 [68]. Preferences against Suno v5 [65] are nearly balanced at 40.4% versus 39.9% (Figure 7). Suno v6 [66] receives more overall preferences, at 59.3% versus 31.6%, despite YuE2 (best-of-8)’s higher SongBench Avg. These percentages retain ties and exclude unable-to-judge responses.

Audio quality is a particular strength. YuE2 (best-of-8) receives an average tie-adjusted audio-quality preference of 58.9% across the six proprietary baselines, with ties split equally between the two systems (CR1 95% CI 55.6–62.2%; $p = 1.70 \times 10^{-6}$). Appendix D.4 reports the complete six-dimension comparisons and their confidence intervals.

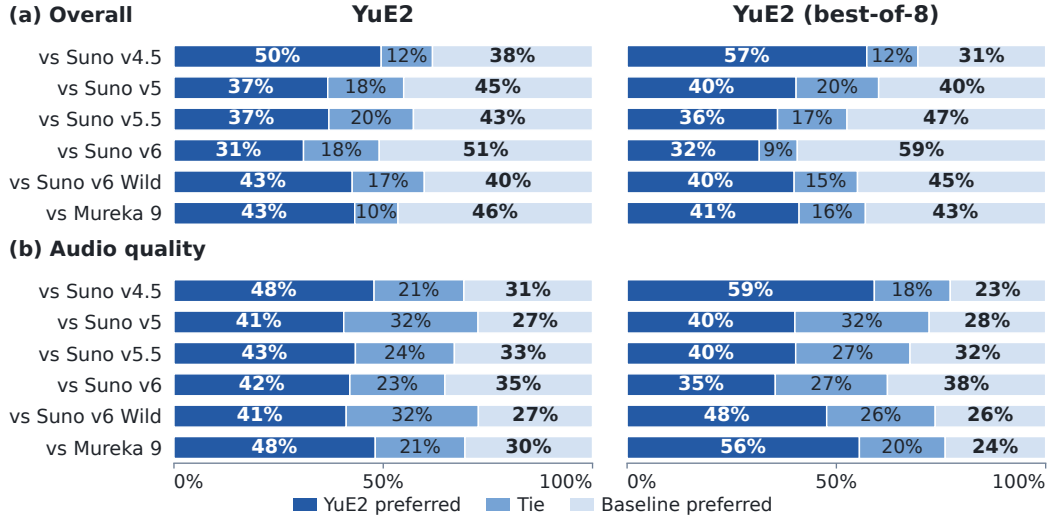


Figure 7: **Expert preferences against proprietary song generators.** Columns show YuE2 and YuE2 (best-of-8), both with symbolic planning. Dark blue favors the YuE2 setting named above the column; light blue favors the baseline named in the row. Baselines are Suno v4.5 [68], v5 [65], v5.5 [62], v6 and v6 Wild [66], and Mureka 9 [51].

We use expert listening as the primary evidence for perceived song quality, with automatic metrics providing a complementary comparison across dimensions. Together, these results establish that YuE2 combines readable, editable composition with frontier full-song quality, competitive with the evaluated proprietary systems.

4.2 SYMBOLIC PLANNING IMPROVES PERCEIVED SONG QUALITY

A score commits the generator to melody and harmony before it produces audio. Whether this early commitment improves the finished song is the central test of symbolic planning. YuE2 supports both generation modes within the same checkpoint, letting us test that choice directly.

Protocol. With planning, the model first generates a melody-and-chord ABC score, then semantic tokens and acoustic latents; without planning, it generates semantic tokens and acoustic latents directly from the text and lyrics. Both conditions are supported by the model’s training mixture, so the comparison requires no retraining. We hold the prompts, lyrics, candidate budget, and waveform decoder fixed. For each of the 192 WSB prompts, we generate two candidates per condition and select the one with lower PER, using the lowest PER from four ASR transcriptions of each candidate. This matches the candidate budget while allowing the planning condition to generate its additional score tokens.

The Arena expert panel evaluates anonymized pairs in randomized A/B order, with at least 30 seconds of listening per song and the consistency checks described in Appendix D.4. The direct plan-

ning comparison contains 212 retained expert evaluations, with 211 judged responses each for overall quality and musicality. For melody and chord progression, four listeners drawn from the same expert pool assess 100 song pairs under the same checkpoint, prompt, candidate-budget, and decoder controls. Each listener assesses 50 pairs, and each pair receives two evaluations from different listeners, yielding 200 judgments per criterion. These pairs also use hidden system identities and randomized presentation order. We exclude unable-to-judge responses and report each criterion’s win/tie/loss shares, retaining ties in each criterion’s denominator.

Results. Experts prefer songs generated with melody-and-chord symbolic planning for both overall quality and musicality (Figure 8). For overall quality, 49.3% of judged responses favor planning and 34.6% favor generation without planning; the rest are ties. The corresponding musicality shares are 45.0% and 29.4%. Both differences are significant in two-sided tests with CR1 standard errors clustered by evaluator and song ($p = 0.0070$ for overall quality and $p = 0.0141$ for musicality; Appendix D.4.4). Preferences also favor planning for melody (44.0% versus 29.5%) and chord progression (39.0% versus 21.0%).

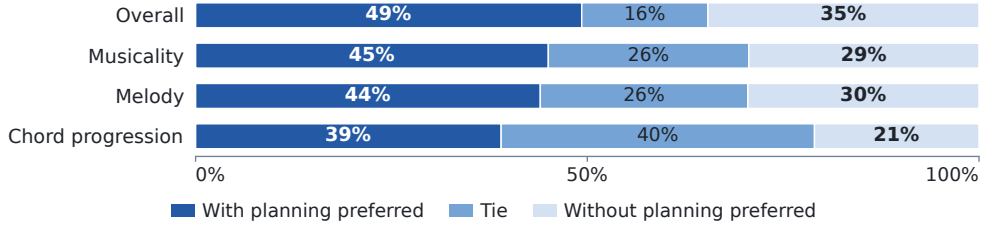


Figure 8: **Symbolic planning improves perceived song quality, melody, and chord progression.** Expert preferences for generation with versus without symbolic planning. Dark blue favors planning, medium blue denotes ties, and light blue favors generation without planning. Appendix D.4 details the listening protocol.

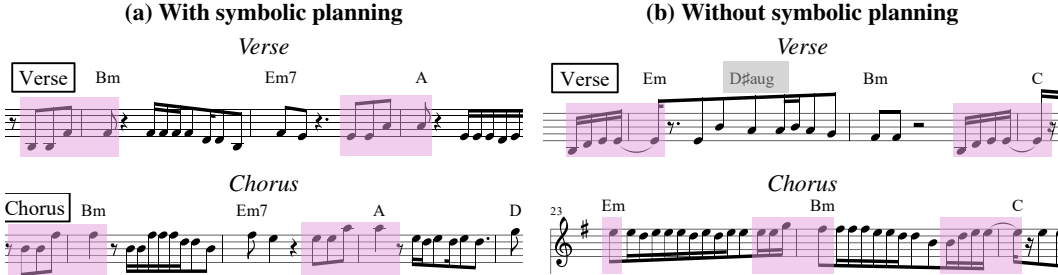


Figure 9: **Planning develops a recurring melodic idea in this paired example.** Both songs use the same prompt and lyrics. Read each staff from left to right; higher notes indicate higher pitch, and letters above the staff name the accompanying chords. Purple marks phrases with repeated lyric openings. **With planning (left)**, the verse varies a low–low–high–high note pattern; the chorus recalls it an octave higher with longer phrase endings, linking the sections through recognizable variation. **Without planning (right)**, the verse repeats the same $B-D-E-E$ pattern, with less variation and a less direct melodic connection to the chorus. Gray marks a chord transition where experts hear the bass descent break, weakening harmonic continuity. The left score starts from the generated plan and the right from a SheetSage2 transcription; experts corrected both against the completed recordings for display, without regenerating audio. Appendix E gives the full scores and further analysis.

Melody and harmony in a paired example. Figure 9 compares expert-corrected score excerpts from songs generated with the same prompt and lyrics. With planning (left), the **purple** motif varies pitch level and intervals while retaining its low–low–high–high contour, then returns an octave higher with longer endings in the chorus. This recognizable variation strengthens thematic continuity and makes the motif more memorable. Without planning (right), the **purple** verse repeats $B-D-E-E$ four times, with little variation and weaker chorus recall.

In the recordings, harmonic motion also sounds more continuous with planning. In the unplanned verse’s gray $Em-D\sharp aug-Bm$ transition, we hear the bass interrupt the descending motion implied by its opening.

4.3 UNIFIED MOT IMPROVES SONG QUALITY OVER SEPARATE LM+DiT

Having established the benefit of symbolic planning, we ask whether semantic-token prediction and acoustic generation should be learned separately or jointly. Many recent song generators pair an autoregressive language model with a separate diffusion Transformer (LM+DiT) [84, 81, 43, 44, 11]. We test whether bringing these stages into YuE2’s unified model improves song quality over this widely used design, with the representations and planning method held fixed.

Protocol. We compare YuE2’s unified AR–NAR Mixture-of-Transformers (MoT) [14] with a separate LM+DiT baseline following Qwen-Music [81]. We match the training-data volume between the two systems. The LM and DiT each have 1.7B parameters (3.4B in total), comparable to the 3.58B MoT. The LM and MoT’s AR component use the same token budget, batch size, and number of training updates. The DiT is trained for 900,000 updates. Both systems use the same semantic tokenizer, acoustic VAE, and melody-and-chord symbolic planning, so the comparison tests unified versus separate generation with the same representations and planning method.

We evaluate both systems on the same 192 WSB prompts and lyrics, generating two candidates per prompt. Each candidate is transcribed four times with ASR, its lowest PER is retained, and the lower-PER candidate is selected. The Arena expert panel compares the resulting anonymized audio pairs under the same randomized presentation, minimum listening duration, and consistency checks as the planning experiment. Experts judge overall quality, musicality, text alignment, audio quality, vocals, and accompaniment separately, with ties and unable-to-judge responses allowed. The direct comparison with planning contains 209 retained expert evaluations, yielding 206–208 judged responses per criterion. Figure percentages include ties and exclude unable-to-judge responses; Appendix D.4.4 reports the criterion-specific counts, CR1 intervals clustered by evaluator and song, and two-sided p -values.

Results. With melody-and-chord symbolic planning in both systems, experts prefer YuE2’s unified MoT to separate LM+DiT. For overall quality, 53.4% of judged responses favor YuE2 and 35.6% favor separate LM+DiT ($p = 0.0084$); the rest are ties. The corresponding audio-quality shares are 48.5% and 29.6% ($p = 3.06 \times 10^{-4}$; Figure 10). Preferences also favor YuE2 for musicality, vocals, and accompaniment; text alignment is nearly balanced at 34.0% versus 33.5%.

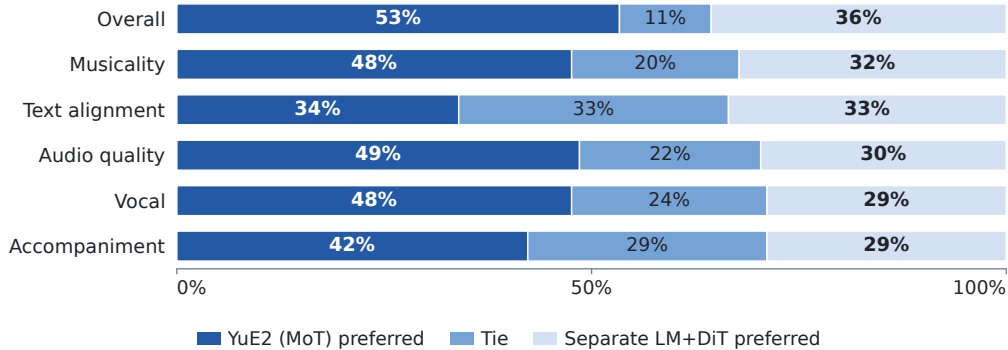


Figure 10: **Experts prefer unified generation on five of six criteria; text alignment is nearly balanced.** YuE2’s Mixture-of-Transformers (MoT) [14] learns semantic-token prediction and acoustic generation in one backbone; the baseline uses a separate language model and diffusion Transformer (LM+DiT) [81]. Both use melody-and-chord planning, the same training-data volume and representations, and the same prompts and two-candidate selection protocol. Dark blue favors YuE2, medium blue denotes ties, and light blue favors LM+DiT. Bars show percentages of judged expert responses (206–208 per criterion), excluding unable-to-judge answers.

5 THE SCORE AS A GENERATIVE INTERFACE

The listening comparisons establish that planning improves perceived song quality. For the score to become an editing interface, its musical decisions must also survive rendering and remain open to revision. We test this connection through score–audio agreement, selective edits, and preservation of work identity across styles. The final case study uses the same interface for iterative editing with a language-model agent.

5.1 THE GENERATED PLAN IS REALIZED IN AUDIO

We test whether YuE2 follows the musical decisions in the score it generates. Given a prompt and lyrics, the model first generates an ABC score and then produces audio conditioned on that score. We pass the generated audio through a frozen SheetSage2 analyzer to recover vocal notes, chords, keys, beats, and section labels. Music experts checked and corrected each transcription. Comparing these events with the generated score measures how closely the recording realizes the planned composition.

Protocol. We evaluate two candidates for each of the 192 WSB prompts, retaining all candidates without quality-based selection. The experiment contains 384 generated scores and 768 recordings, including two recordings per prompt generated without a score. Every recording is transcribed in the same way. We compare each score with three recordings: its own realization (*Corresponding*); the audio generated from the other score under the same prompt and lyrics (*Mismatched*); and the corresponding recording generated directly without a score (*Without score*). The two controls test whether agreement is specific to the particular composition in the score when the text conditions are held fixed.

Evaluation. We evaluate the transcription against the generated score, using the notated pitches and tempo as targets. Melody and chord similarities each use one minus normalized edit distance between the notated and transcribed sequences, using sounding pitches for melody and root-and-triad labels for chords. Key evaluation reports exact tonic-and-mode agreement and weighted agreement that gives partial credit for related keys. Form similarity compares section roles, order, and melodic and harmonic content.

Rhythm F1 measures agreement between consecutive vocal onset intervals in beats. For tempo, we report the absolute log ratio of estimated to notated BPM and the fraction of estimates within 8% of the target. Section-boundary F1 matches section starts within two beats after a single global time shift derived from melody matches. Appendix D.1 gives the full metric definitions.

Table 3: **Score–audio consistency on WSB.** Mismatched audio uses the other score for the same prompt. Rhythm reports vocal onset-interval F1. Values are prompt means. Tempo accuracy allows 8% error; lower log error is better.

Audio	Melody	Chords	Rhythm	Key		Tempo		Form	
	Seq. ↑	Seq. ↑	F1 ↑	Exact ↑	Wtd. ↑	Log err. ↓	Acc. ↑	Content ↑	Bound. ↑
Corresponding	0.9464	0.9246	0.9396	0.9307	0.9515	0.0142	0.9869	0.7799	0.8724
Mismatched	0.2160	0.2473	0.6119	0.3004	0.3629	0.1131	0.6806	0.1547	0.1454
Without score	0.2055	0.1897	0.5875	0.2234	0.2831	0.1468	0.5366	0.1236	0.1245

Content and timing fidelity. Table 3 reports prompt means. Depending on the reference content available for each metric, 378–380 scores from 191 prompts are evaluable. Corresponding recordings achieve melody and chord sequence similarities of 0.9464 and 0.9246, compared with 0.2160 and 0.2473 for mismatched recordings. The large gap shows that the recovered content identifies the particular planned composition, beyond agreement with the shared text prompt. Exact key agreement is 0.9307, and tempo is within 8% of the target for a prompt-averaged fraction of 0.9869. Vocal onset-interval F1 is 0.9396. Form similarity is 0.7799, reflecting agreement in section role,

order, and content; section-boundary F1 is 0.8724 after a global start adjustment. The corresponding recordings retain the planned pitch content while following its rhythm and global tempo. In the WSB two-candidate assignment task, melody and chord similarity each yield a higher total for the correct score–audio pairings than for the swapped pairings in every evaluable case (100% assignment accuracy; Appendix D.1).

The melody and harmony advantage over both controls persists when each score–audio comparison allows one song-wide pitch shift (Appendix D.1). The gap therefore extends beyond differences in global key and register.

5.2 CONTROLLED SCORE EDITING

Although YuE2 regenerates the entire song after a score edit, the requested change can be local. We test edit adherence and preservation together: whether the revised passage follows its new score while unedited melody and harmony remain faithful to the source composition.

Protocol. We take the first generated score for each of the 192 WildSongBench prompts and evaluate melody, harmony, rhythm, key, and tempo edits separately. Melody and rhythm edits affect up to four bars of the first chorus; harmony substitutions extend to matching phrases. Key and tempo changes apply to the whole score. Rhythm edits exchange adjacent quarter- and eighth-note durations, preserving pitches and each pair’s total duration. We regenerate complete songs from the edited scores and unedited controls, holding lyrics, style, model, decoder, and sampling settings fixed. Two seeds per applicable condition yield 3,844 recordings, and all outputs are retained.

Evaluation. We measure harmony adherence and content preservation from SheetSage2-AR transcriptions, and melody, rhythm, and key adherence from SheetSage2-Prober transcriptions. Music experts review and correct the transcriptions of generated songs for editing evaluation. Tempo is estimated directly from audio using madmom [6]. Table 4 separates edit adherence from content preservation. Melody adherence is exact target-pitch accuracy on edited notes. Harmony adherence is duration-weighted root-and-triad agreement in the edited chorus window. Rhythm adherence measures relative onset intervals after normalizing local offset and tempo. An interval must be within 1/8 beat of the edited target and closer to it than to the original timing. Weighted key agreement gives full credit for the target tonic and mode and partial credit for related keys. Tempo Acc2 [57] accepts BPM estimates within 4% of the target multiplied by any of $\{1/3, 1/2, 1, 2, 3\}$.

Content preservation measures agreement with unedited melody and harmony in the source score. We average seeds and edit strengths within each song, then weight songs equally.

Results. Local edits change the requested content while largely preserving unedited melody and harmony. Harmony edits attain 79.54% target-chord agreement while retaining 93.43% full-song vocal melody agreement. Melody edits attain 84.17% exact target-pitch accuracy, and rhythm exchanges attain 73.43% relative onset accuracy. Across these three local edits, agreement with unedited melody and harmony ranges from 90.16% to 94.34%. For global changes, key editing achieves a weighted score of 90.58/100, and tempo editing reaches 95.68% Acc2.

SongBench quality remains close to the corresponding unedited controls: 6.64–6.73 across melody, harmony, rhythm, and key edits, versus 6.70 before melody, harmony, and key edits and 6.73 before rhythm edits. Appendix D.2 gives the full protocol, metric definitions, and quality and lyric-retention results.

Table 4: **Score editing on WildSongBench.** Higher is better for all scores (0–100).

(a) Edit adherence			(b) Content preservation		
Edit	Metric	Score $\uparrow$	Edit	Melody $\uparrow$	Harmony $\uparrow$
Melody	Pitch accuracy	84.17	Melody	93.10	94.05
Harmony	Chord agreement	79.54	Harmony	93.43	90.16
Rhythm	Relative onset accuracy	73.43	Rhythm	94.34	93.08
Key	Weighted key score [87]	90.58	Agreement with unchanged score content.		
Tempo	Acc2 [57]	95.68			

Appendix D.2 gives the protocol and metric definitions.

5.3 ZERO-SHOT COVER GENERATION PRESERVES WORK IDENTITY

Cover generation tests whether the learned score-to-audio mapping extends beyond the composition–style pairings seen during training: a supplied composition must remain recognizable in a different style. We evaluate this transfer with the general YuE2 checkpoint, without original–cover training pairs or cover-specific fine-tuning. Our *zero-shot cover generation* evaluation uses 948 works from the official SHS100K test split [53]. These works, including their alternate performances, are absent from YuE2’s training corpus.⁶

Protocol. For each work, we use one source recording. SheetSage2 transcribes it into a score, and automatic speech recognition recovers its lyrics. We automatically derive two contrasting style descriptions from two other performances of the same work. YuE2 generates a new recording conditioned on the fixed source score, the source lyrics, and one target-style description. Two seeds per target style yield four recordings per work and 3,792 outputs per method. We retain all outputs without candidate selection.

We compare the full score with a version without chords and with no score, holding the YuE2 checkpoint, lyrics, style requests, and audio decoder fixed. Rebuilding the score without chords also changes its serialization.

We also evaluate SongEcho [45] and ACE-Step 1.5 [23]. Both systems receive the source audio, lyrics, and target-style descriptions through their native cover-generation interfaces.

Evaluation. Each generated cover serves as a query for retrieving other performances of its source work. CLEWS [59] and Discogs-VINet [4] rank the same gallery of 10,546 recordings from 1,692 works. We exclude the exact source recording for each query and treat the remaining recordings of the same work as correct matches. For each retrieval model, we report mean average precision (mAP), mean reciprocal rank (MRR), and Hit@1/Hit@5. Hit@ k is the percentage of queries with at least one correct match among the top k results. Higher retrieval scores indicate stronger preservation of work identity. Appendix D.3 gives the full protocol and metric definitions.

Results. Without cover-specific training, YuE2 with the full score leads both evaluated cover systems on all eight retrieval measures, production quality, and Musicality (Table 5). CLEWS and Discogs-VINet mAP reach 0.647 and 0.288, respectively, compared with 0.419 and 0.122 for SongEcho, the stronger retrieval baseline. Within YuE2, removing chords lowers the two mAP scores to 0.598 and 0.179; removing the entire score lowers them to 0.006 and 0.004.

MuQ-MuLan [91] and Q3O [80] measure alignment with the requested target style; AudioBox Aesthetics [70] and SongBench [77] measure production quality (PQ) and Musicality. Within YuE2, relaxing score conditioning raises alignment and quality scores but reduces work-identity retrieval. Musicality rises from 5.104 with the full score to 5.490 without chords and 5.691 without a score; PQ follows the same ordering. MuLan and SongBench cover all 3,792 outputs per method; Q3O covers a matched subset of 3,286 outputs per method (Appendix D.3).

⁶We verified this using CLEWS [59] and Discogs-VINet [4] to check for work-level overlap with the training corpus.

Table 5: **YuE2 with full scores leads both evaluated cover systems on all eight retrieval measures without cover-specific training.** SHS100K [53] evaluation across 948 works, with 3,792 outputs per method and no candidate selection. These works are absent from YuE2’s training corpus. Each query ranks 10,545 recordings after excluding its source. All metrics are higher-is-better.

Method	CLEWS				Discogs-VINet				Alignment		Quality	
	mAP ↑	MRR ↑	Hit@1 ↑	Hit@5 ↑	mAP ↑	MRR ↑	Hit@1 ↑	Hit@5 ↑	MuLan ↑	Q3O ↑	PQ ↑	Mus. ↑
SongEcho	0.419	0.536	48.4	58.8	0.122	0.227	16.6	28.4	0.366	4.474	6.862	3.286
ACE-Step 1.5	0.024	0.036	2.4	4.1	0.006	0.014	0.6	1.4	0.166	4.190	6.918	3.689
YuE2 (full score)	0.647	0.748	71.3	78.6	0.288	0.438	37.5	50.3	0.382	4.273	8.044	5.104
Without chords	<u>0.598</u>	<u>0.715</u>	<u>67.3</u>	<u>76.1</u>	<u>0.179</u>	<u>0.304</u>	<u>23.6</u>	<u>36.6</u>	<u>0.417</u>	<u>4.482</u>	<u>8.117</u>	<u>5.490</u>
Without score	0.006	0.008	0.3	0.9	0.004	0.008	0.2	0.8	0.474	4.837	8.186	5.691

Hit@ k is in percent; Q3O ranges from 0 to 5. PQ: AudioBox production quality; Mus.: SongBench Musicality. **Bold:** column best; underline: second best.

Our qualitative assessment with music experts links this pattern to harmony’s role in musical style: retaining source harmony favors fidelity to the original work, while relaxing it allows greater adaptation to the target style. The cover task renders a fixed source transcription in a contrasting style; in the planning preference study, the model composes a score for the requested style.

These cover results demonstrate transfer of the same score interface to unseen works without cover-specific training. They also motivate two ways to adapt a source composition: relaxing harmonic conditioning during cover generation, or explicitly revising the score through agentic music editing.

5.4 AGENTIC MUSIC EDITING: A CASE STUDY

Adapting a song to a new style can require revising the composition itself. YuE2’s readable ABC score gives an external language-model agent direct access to melody, harmony, and form. In *agentic music editing*, a user shapes the song through musical dialogue: the agent turns requests into revisions of the score, style description, and lyrics, while the same YuE2 model renders each version as a complete song.

*The Last Train*⁷ begins as Mandarin pop and is reworked into English jazz through collaboration between a researcher and a language-model agent. Nine curated steps and fourteen rendered versions capture changes to harmony, melodic phrasing, and lyrics, including a saxophone interlude built around *Twinkle, Twinkle, Little Star*. The researcher steers the musical direction and chooses the final recording by listening; retained alternatives and corrections show how the composition develops through feedback. The score is both the plan that guides YuE2 and the composition that the user and agent continue to write together.

6 MUSIC UNDERSTANDING AND FULL-SONG TRANSCRIPTION

MERT2 leads 14 of 15 MARBLE metrics, and SheetSage2-AR leads 12 of 15 benchmark–metric pairs in full-song transcription. At 375 bits/s, the MERT2 semantic tokenizer leads five of six metrics among the three evaluated quantized representations. The tokenizer and SheetSage2 provide semantic tokens and symbolic annotations for YuE2, respectively.

6.1 MERT2 LEADS MUSIC-UNDERSTANDING BENCHMARKS

A full-song representation must capture long-range context without losing local pitch and rhythm. We test whether extending pretraining from excerpts to complete songs preserves this musical information by comparing MERT2-30s with its full-song continuation, MERT2-FS, on nine MARBLE tasks [87]. We freeze each encoder and train task-specific prediction heads on its features. MERT2-30s is pretrained on 30-second excerpts; MERT2-FS continues pretraining on complete recordings

⁷Editing conversation, scores, and audio: <https://map-yue2.github.io/#agentic-music-editing>.

lasting 30–360 seconds. Downstream tagging uses MARBLE’s standard 30-second task excerpts. Exact probe settings are given in Appendix B.

Table 6: MERT2-30s leads on 14 of 15 MARBLE [87] metrics against the listed external baselines. Baseline scores are from Tables III and V of Gu et al. [26] (arXiv v3). AudioMAE++, MATPAC++, and M2D-Large are that study’s retrained models; the remaining external baselines use its evaluations of released checkpoints or its own PupuJEPA models. Results cover MagnaTagATune (MTT) [42], GiantSteps [39], GTZAN [71], EmoMusic [63], and MTG-Jamendo [7]. MERT2-FS (full-song) continues pretraining on complete recordings lasting 30–360 seconds.

Dataset Task		MTT Tagging		GiantSteps Key	GTZAN Genre Beat		EmoMusic Emotion	
Model	# Params	ROC $\uparrow$	AP $\uparrow$	Acc. $\uparrow$	Acc. $\uparrow$	F1 ^{beat} $\uparrow$	R_V^2 $\uparrow$	R_A^2 $\uparrow$
MERT-Large [46]	330M	90.6	37.9	64.1	77.6	86.8	56.7	76.1
Dasheng-1.2B [15]	1.2B	91.5	40.4	58.0	81.4	87.7	57.4	75.0
MuQ [91]	310M	90.5	38.5	63.2	83.8	90.1	58.3	76.4
MusicFM [76]	330M	90.9	38.3	63.0	84.1	90.2	57.2	74.4
AudioMAE++ [82]	307M	91.2	39.5	61.7	80.3	90.0	59.0	75.7
MATPAC++ [55]	307M	90.6	38.2	63.7	81.4	90.1	57.8	74.7
M2D-Large [26]	307M	91.0	39.2	65.0	83.8	90.0	57.4	74.8
PupuJEPA-Large [26]	307M	91.7	40.8	66.1	86.9	91.0	62.5	76.8
PupuJEPA-Huge [26]	632M	91.3	39.7	64.8	85.9	90.5	62.0	<u>78.5</u>
MERT2-30s	632M	91.91	41.29	<u>66.97</u>	91.72	<u>90.59</u>	<u>63.23</u>	80.01
MERT2-FS	632M	<u>91.74</u>	<u>41.20</u>	67.05	<u>90.69</u>	<u>90.57</u>	63.52	78.14

Dataset Task		MTG-Jamendo							
		Instrument		Mood / theme		Genre		Top 50	
Model	# Params	ROC $\uparrow$	AP $\uparrow$	ROC $\uparrow$	AP $\uparrow$	ROC $\uparrow$	AP $\uparrow$	ROC $\uparrow$	AP $\uparrow$
MERT-Large [46]	330M	75.5	18.8	75.3	13.5	86.1	18.0	82.6	29.1
Dasheng-1.2B [15]	1.2B	75.0	19.0	76.1	15.5	85.5	18.8	82.4	29.6
MuQ [91]	310M	74.8	19.1	73.7	13.2	85.4	19.1	83.0	30.2
MusicFM [76]	330M	74.6	18.5	74.9	14.1	85.3	19.4	81.9	29.7
AudioMAE++ [82]	307M	77.1	19.9	75.6	14.0	86.3	18.9	83.1	31.1
MATPAC++ [55]	307M	77.2	19.7	75.1	14.1	85.7	19.6	82.5	30.2
M2D-Large [26]	307M	76.6	19.3	74.6	14.3	85.5	19.2	82.5	29.6
PupuJEPA-Large [26]	307M	<u>78.4</u>	21.2	76.2	15.3	86.1	20.1	82.8	30.5
PupuJEPA-Huge [26]	632M	77.6	20.5	75.9	14.7	85.9	20.1	83.1	30.7
MERT2-30s	632M	80.27	<u>22.89</u>	79.44	16.68	88.01	21.22	84.18	32.17
MERT2-FS	632M	80.27	23.51	<u>78.74</u>	<u>15.74</u>	<u>87.98</u>	<u>20.66</u>	<u>84.13</u>	<u>31.62</u>

Naming note. The authors renamed PupuJEPA to PupuM2D after the paper’s initial release. We retain PupuJEPA.

Table 6 compares MERT2 with MERT [46], Dasheng [15], MuQ [91], MusicFM [76], AudioMAE++ [82], MATPAC++ [55], and the M2D-Large and PupuJEPA models reported by Gu et al. [26]. MERT2 establishes a new state of the art on MARBLE. Across nine tasks, MERT2-30s exceeds every external baseline on 14 of 15 metrics. Its full-song continuation, MERT2-FS, exceeds every external baseline on 13 of 15 metrics and remains within 1.87 points of MERT2-30s across all 15. It achieves the highest scores on GiantSteps [39] key, EmoMusic valence, and MTG-Jamendo [7] Instrument AP, and ties MERT2-30s on Instrument ROC. MERT2 achieves leading results in music understanding, with strong performance retained after full-song adaptation.

6.2 MERT2 RETAINS STRONG MUSICAL STRUCTURE AT 375 BITS/S

The preceding results establish MERT2’s strong music-understanding performance with full-context representations. For YuE2, the critical question is whether this musical structure can survive a discrete bottleneck of just 375 bits/s. We test this with matched MARBLE probes for genre, emotion, key, music tagging, and beat tracking. The probes use frozen 1,024-dimensional semantic-tokenizer

states before and after vector quantization (pre-VQ and post-VQ) [72]. We also compare the post-VQ state with LeVo 2 [44] and EnCodec [13] at their generation-time quantized outputs. Every representation uses the same split, task-decoder architecture apart from its input width, optimization recipe, and seed within a task; Appendix B.2 defines the probe points, native rates, and temporal resampling.

Table 7: **At 375 bits/s, the MERT2 tokenizer leads five of six metrics among the evaluated quantized representations.** Scores are multiplied by 100 (higher is better), with the best post-VQ result per column in bold. Emotion reports the mean R^2 across arousal and valence. Rate, feature dimension, and nominal bitrate characterize each representation; external controls are LeVo 2 [44] and EnCodec [13].

Representation	Token stream			Genre	Emotion	Key	MTT		Beat
	Rate (Hz)	Dim.	kb/s	Acc. $\uparrow$	R^2 $\uparrow$	Score $\uparrow$	AUROC $\uparrow$	AP $\uparrow$	F1 $\uparrow$
MERT2 tokenizer pre-VQ	25	1024	–	84.14	66.19	56.94	91.60	40.02	89.81
MERT2 tokenizer post-VQ	25	1024	0.375	65.86	54.49	52.20	90.06	35.65	86.38
LeVo 2 post-VQ	25	1024	0.350	44.83	30.17	61.23	86.72	30.79	79.72
EnCodec post-VQ	75	128	1.500	31.72	26.51	15.76	80.84	22.32	76.50

At 375 bits/s, the MERT2 tokenizer leads the two evaluated quantized alternatives on five of six frozen-probe metrics (Table 7). Its beat F1 reaches 86.38, compared with 79.72 for LeVo 2 and 76.50 for EnCodec.

Across the six metrics, post-VQ probe scores are 78.3–98.3% of the corresponding pre-VQ scores. The semantic token stream supports strong music understanding at 375 bits/s.

6.3 SHEETSAGE2-AR LEAD-SHEET TRANSCRIPTION

SheetSage2-AR learns full-song transcription exclusively from labels generated by SheetSage2-Prober. We test whether this single model can surpass its label generator and compete with task-specific systems while jointly transcribing melody, chords, key, beats, downbeats, and structure. Table 8 evaluates SheetSage2-AR against SheetSage1 [16], madmom [6], and the task-specific Beat This! [21], S-KEY [40], ChordFormer [2], and SongFormer [28]. The comparison spans GTZAN [71], osu2017 [33], GiantSteps [39], Chords1217 [31], Jazz Audio-Aligned Harmony (JAAH) [18], HarmonixSet [52], RWC-Pop [24], and Rock Corpus (RS 200) [69]. The table and Appendix B.3 state the protocol differences.

A single SheetSage2-AR model achieves the highest scores on 12 of 15 benchmark–metric pairs in this comparison. SheetSage2-AR substantially improves over SheetSage1 in melody, chord, key, and downbeat estimation, and surpasses task-specific models on several benchmarks.

Table 8: **SheetSage2-AR: one model for full-song lead-sheet analysis.** The comparison includes SheetSage1 [16], madmom [6], and task-specific systems cited beside their scores. SheetSage2-AR achieves the highest scores on 12 of 15 benchmark–metric pairs. Scores are percentages; higher is better.

Task	Benchmark	Metric	SheetSage1	Madmom	Prior specialist	SheetSage2-AR
Beat	GTZAN osu2017	F1 ↑	86.07	86.07	89.01 [21]	86.27
			91.80	91.80	89.19 [21]	93.01
Downbeat	GTZAN osu2017	F1 ↑	64.65	64.65	78.28 [21]	80.45
			83.47	83.47	85.90 [21]	92.90
Key	GiantSteps GTZAN	Score ↑	43.89	74.62	72.09 [40]	77.73
			54.56	72.05	74.43 [40]	75.77
Chord	osu2017	Maj/min ↑	79.82	77.42	86.55 [2]	90.08
	Chords1217		72.98	83.52*	83.94 [†] [2]	83.81
	JAAH		39.77	51.24	59.45 [2]	64.50
Structure	HarmonixSet	Accuracy ↑	—	—	80.03 [28]	80.51
		F1 (0.5 s) ↑	—	—	70.63 [28]	67.96
		F1 (3 s) ↑	—	—	79.50 [28]	82.86
Melody	RWC-Pop	Vocal F1 ↑	62.71	—	62.71 [16]	82.51
		Full F1 ↑	64.02	—	64.02 [16]	75.29
	Rock Corpus	Vocal F1 ↑	49.19	—	49.19 [16]	67.08

Bold marks the best value for each benchmark–metric pair; an em dash denotes unavailable results. Structure F1 measures section boundaries at the stated tolerance. Melody F1 uses a 50-ms onset tolerance and pitch classes. SheetSage1 rhythm entries reuse Madmom outputs; the melody specialist repeats SheetSage1. Beat This! scores average three released training seeds. *The madmom chord model was trained on data that include Chords1217. [†]ChordFormer uses five-fold evaluation, with one held-out model per track. Its original scorer pools metric-eligible durations; we recompute per-track recall and weight it by each reference span to match the other models’ scoring protocol. SheetSage1 and SheetSage2-AR each evaluate one fixed checkpoint on all 1,217 tracks. Appendix B.3 gives the evaluation protocols and score sources.

On RWC-Pop, vocal melody pitch-class F1 rises from 62.71 to 82.51, a gain of 19.80 percentage points over SheetSage1. On Rock Corpus, vocal melody pitch-class F1 rises from 49.19 to 67.08 under the same 50-ms onset criterion. On all 113 JAAH recordings, SheetSage2-AR reaches 64.50% maj/min recall, exceeding ChordFormer by 5.06 percentage points (Appendix B.3).

Trained exclusively on labels generated by SheetSage2-Prober, SheetSage2-AR surpasses its label generator on 10 of 15 benchmark–metric pairs under the same evaluation protocol (Table 11).

7 RELATED WORK

Audio generation and implicit composition. MusicLM [1] and MusicGen [10] model text-conditioned music through discrete audio representations; MusicFlow [54] uses cascaded flow-matching networks for semantic and acoustic features. YuE [88] scales autoregressive generation to full songs; SongBloom [83] and LeVo 2 [44] combine autoregressive modeling with diffusion, while DiffRhythm 2 [36] uses block flow matching. HeartMuLa [84] and Muse [32] further develop open song generation with style control. These systems model musical structure in audio representations, without exposing a generated score in which users can inspect and revise melody, harmony, and form together. YuE2 makes this composition an explicit intermediate while achieving frontier full-song quality in automatic benchmarks and expert listening.

Symbolic composition and full-song realization. Music Transformer [29] and SymPAC [8] expose musical decisions but generate symbolic sequences rather than recordings. MIDI-DDSP [79] connects scores to expressive instrumental performance; BACH [74] extends symbolic generation to songs, with external tools synthesizing vocals and accompaniment. Thus, generating notation alone leaves its realization as a complete song to a separate system. Seed-Music [5] describes a pro-

proprietary lead-sheet-to-song pipeline with limited implementation details and no publicly available implementation. YuE2 jointly models readable scores, semantic music tokens, and acoustic latents within one publicly available generator. The semantic and acoustic representations carry the performance detail that a sparse score leaves unspecified. The same checkpoint supports song creation, score-based editing, and zero-shot cover generation.

Musical control and planning. Music ControlNet [78] conditions generation on time-varying melody, dynamics, and rhythm. Joint Audio and Symbolic Conditioning (JASCO) [67] combines textual prompts with local symbolic and audio controls. These methods follow supplied controls; composing those controls remains outside the generator. Planning methods address this step with different representations: MusiCoT [41] generates quantized audio–text embeddings before audio tokens; ACE-Step 1.5 [23] plans tempo, key, duration, and structure. These plans do not jointly specify melody, harmony, rhythm, and form in readable notation. YuE2 generates such a score for the full song, allowing people and external language-model agents to revise the composition before rendering. Matched listening comparisons show that this explicit composition improves perceived melody, harmony, and overall song quality. Qwen-Music [81] conditions covers on relative vocal pitch contours from a reference recording. SongEcho [45] learns dedicated melody-conditioning modules; YuE2 reuses its score-to-audio mapping without cover-specific training.

Representations for analysis and generation. MERT [46], MusicFM [76], and MuQ [91] learn transferable music representations through self-supervision. Strong analysis features alone do not supply a compact discrete sequence suited to autoregressive generation. MERT2 addresses both requirements from a shared pretrained encoder, using Multi-View Target Synthesis to construct targets constrained by complementary frozen encoder views. Its full-context branch supports transcription; a separate branch undergoes causal-attention adaptation and quantization to supply generation tokens. Its analysis representations lead 14 of 15 MARBLE [87] metrics in our comparison.

Transcription as supervision for generation. Multi-Task Multitrack Music Transcription (MT3) [22] predicts multitrack notes, while SheetSage [16] combines melody transcription with beat, key, and chord analysis to produce lead sheets. Beat This! [21], ChordFormer [2], and SongFormer [28] specialize in beat, chord, and structure analysis, respectively. These approaches do not jointly predict notes, harmony, metrical organization, and section structure across a full song. Combining their outputs also requires reconciling attribute vocabularies and timing conventions. SheetSage2-AR jointly predicts melody, chords, key, beats, downbeats, and structure in one full-song sequence, with the highest scores on 12 of 15 benchmark–metric pairs in our comparison. SheetSage2 constructs aligned symbolic targets under a shared vocabulary and timing convention. Together with MERT2 semantic tokens, these targets make the composition-to-audio hierarchy learnable from recordings without pre-existing aligned scores.

8 CONCLUSION

YuE2 unifies readable composition and full-song audio generation at frontier quality. Experts prefer symbolic planning for overall quality and musicality, and favor the unified MoT over separate LM+DiT under melody-and-chord planning. Overall expert preferences favor best-of-8 over Suno v4.5 and are nearly balanced against Suno v5 and Mureka 9. Audio-quality preferences favor best-of-8 on average across six proprietary systems. On WildSongBench, YuE2 leads all evaluated public baselines on SongBench Global Avg under the standard two-candidate protocol; best-of-8 achieves the highest observed mean across all evaluated systems. MERT2 sets a new state of the art on 14 of 15 MARBLE metrics, and SheetSage2-AR leads 12 of 15 benchmark–metric pairs in our full-song transcription comparison. Together, they supply the semantic and symbolic supervision that makes YuE2 trainable from recordings at scale.

The score provides an interface for composition and intervention: melody and harmony survive rendering, and local edits selectively change musical content. Without cover-specific training, the same checkpoint uses full scores to generate covers of unseen works, surpassing both evaluated cover systems on all eight work-identity retrieval measures, AudioBox production quality, and SongBench Musicality. Explicit composition gives a full-song generator musical decisions that can be inspected and revised before they become sound.

CONTRIBUTORS AND ACKNOWLEDGEMENT

Core Contributors*

Ruibin Yuan

Lead, Model, Data, Eval

The Hong Kong University of Science and Technology
ACE Studio
Multimodal Art Projection
ryuanab@connect.ust.hk
ruibiny@acestudio.ai

Jiahao Pan

Model, Data

The Hong Kong University of Science and Technology
Multimodal Art Projection
fengshicherish@gmail.com

Junyan Jiang

Model

New York University
Mohamed bin Zayed University of Artificial Intelligence
ACE Studio
Multimodal Art Projection
jj2731@nyu.edu

Zhiyue Wu

Model

StepFun
Multimodal Art Projection
zhiyuewu70@gmail.com

Ziya Zhou

Eval

The Hong Kong University of Science and Technology
Multimodal Art Projection
zzhoucp@connect.ust.hk

Jiankai Sun

Compute

Stanford University
Multimodal Art Projection
jksun@stanford.edu

Yizhi Li

Compute

Multimodal Art Projection
yizhi.li-2@manchester.ac.uk

Ge Zhang

Data

Tokenwave.AI
Multimodal Art Projection
gezhang@umich.edu

Corresponding Authors

Wei Xue

The Hong Kong University of Science and Technology
weixue@ust.hk

Yike Guo

The Hong Kong University of Science and Technology
yikeguo@ust.hk

Contributors

Yicheng Gu

Aalto University

Zeyue Tian

The Hong Kong University of Science and Technology
NOIZ.ai

Junyu Dai

Multimodal Art Projection

Hanfeng Lin

Muse.ai
The Hong Kong University of Science and Technology

Kai Li

Tsinghua University

Shangda Wu

Multimodal Art Projection

Xuanjie Liu

Mohamed bin Zayed University of Artificial Intelligence

Jiaming Wang

NOIZ.ai

Zihan Liu

Multimodal Art Projection

Yue Wang

Soochow University

Yinghao Ma

Multimodal Art Projection

Hanzhi Yin

Multimodal Art Projection

Kangrui Chen

The Hong Kong University of Science and Technology

Xinyue Zhang

Multimodal Art Projection

Ziyang Ma

Shanghai Jiao Tong University

Mengqi Liao

Hillhouse Investment

Hejia Zhao

The University of Hong Kong

Guowei Huang

Multimodal Art Projection

Chao Yan

StepFun

Lei Ke

NOIZ.ai

Jianwei Yu

Multimodal Art Projection

Bei Liu

The Hong Kong University of Science and Technology

Joe Guo

ACE Studio

Liumeng Xue

Nanjing University

Gus Xia

Mohamed bin Zayed University of Artificial Intelligence
New York University

*Equal contribution.

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A WILDSongBENCH EVALUATION DETAILS

A.1 QWEN3-OMNI PROMPT ALIGNMENT (Q3O)

We use Q3O to denote our Qwen3-Omni-based prompt-alignment score. It measures how well a generated song follows the musical requirements in its WildSongBench request. We evaluate seven dimensions: vocal timbre, vocal setup (solo, duet, group, gender, or instrumental), genre, mood, instrumentation and production, tempo, and key. We use Qwen3-Omni-30B-A3B-Instruct [80] in two passes. A text-only pass assigns each dimension an importance weight $w_{i,d} \in \{0, \dots, 5\}$; zero means that the request does not specify that dimension. For each prompt, the same dimension weights are used across all evaluated systems and sampling settings. An audio pass then assigns each active dimension a match score $s_{i,d} \in \{0, \dots, 5\}$, where five is a strong match and zero is a clear miss.

For request i , we compute

$$C_i = \frac{\sum_d w_{i,d} s_{i,d}}{\sum_d w_{i,d}}, \quad \text{Q3O} = \frac{1}{N} \sum_{i=1}^N C_i. \quad (9)$$

An active key term uses the standard Music Information Retrieval Evaluation eXchange (MIREX) weighted agreement between the requested key and a madmom CNN estimate [6], rescaled from $[0, 1]$ to $[0, 5]$. The reported system score averages all $N = 192$ requests.

Audio-scoring prompt. For reproducibility, the exact instruction and user-message template are shown below. Angle-bracketed fields are filled directly from each WSB record; DIMENSIONS and OUTPUT_SCHEMA contain only dimensions with positive prompt weight.

Q3O: exact audio-scoring prompt

SYSTEM MESSAGE

You are a strict music prompt-control evaluator.

Evaluate whether the provided audio satisfies the user’s open-format music request. Use only the user’s raw prompt, style tags, and raw genre as the target.

Do not use any standardized or annotated prompt fields as ground truth.

Return JSON only. Do not wrap it in markdown.

Score each dimension independently:

- score: integer 0-5. 5 means the audio strongly matches the user’s request for this dimension. 0 means it clearly contradicts or misses the request.
- reason: short reason. Mention the requested target and the audible evidence.

Evaluate only the dimensions listed below. Do not add extra dimensions.

Dimensions:

{{DIMENSIONS}}

Output schema:

{{OUTPUT_SCHEMA}}

Q3O: exact audio-scoring prompt (continued)**USER MESSAGE TEMPLATE · AUDIO ATTACHED**

Evaluate this generated song against the following open-format user request.

Title: <TITLE>
 Language: <LANGUAGE>
 Raw genre: <RAW_GENRE>
 Style tags: <STYLE_TAGS>

Raw user prompt:
 <RAW_USER_PROMPT>

A.2 PRODUCTION QUALITY AND AUDIO-TEXT ALIGNMENT

Each additional metric is evaluated on the same selected audio as the other WSB metrics, with complete coverage of 192 songs per system. System scores are arithmetic means over songs. AudioBox Aesthetics [70] estimates production quality (PQ), including clarity, fidelity and spatial presentation. Its full-song score uses length-weighted 10-s windows.

For MuQ-MuLan [91] and AllMusicCaps [3], the text target is the same standardized style description for each prompt, containing genre, instrumentation, mood and vocal attributes but no lyrics. Thus these scores measure alignment with the style annotation. They complement Q3O, which evaluates alignment with the original user request. Audio is decoded to 24-kHz mono. MuQ-MuLan uses its serial full-song inference with 10-s clips. AllMusicCaps uses the released model with a trained text encoder and SigReg regularization: it averages embeddings over complete, non-overlapping 10-s audio clips, normalizes the mean and text embeddings, then computes cosine similarity. Audio shorter than 10 s is zero-padded; an incomplete trailing clip in longer audio is omitted.

A.3 SELECTION AND EVALUATOR REUSE

Both YuE2 settings use melody-and-chord symbolic planning. For every system, the standard comparison selects the lower-PER output from two candidates per prompt. Candidate budgets count outputs available for selection. For YuE2, the publicly released baselines, and Suno v6 and v6 Wild [66], each candidate’s PER is the lowest across four ASR transcriptions. MiniMax Music 3 [50] uses its official caption rewriter; MM Music abbreviates MiniMax Music in the tables.

For best-of-8, we remove candidates dominated jointly in SongBench Musicality, Q3O and PER, retain candidates within 0.10 of the highest Musicality, then within 0.10 of the highest remaining Q3O, and select the lowest PER. Reported best-of-8 scores evaluate the selected recordings.

HeartMuLa uses SongEval to filter supervised fine-tuning data and, with AudioBox, to construct a DPO preference set; its final model merge includes the resulting DPO model [84]. The tables flag this evaluator reuse and include all systems in the per-column rankings.

A.4 CALCULATION OF THE QUALITY AND ALIGNMENT INDICES

Figure 1 maps each system to a song quality index (horizontal axis) and a text alignment index (vertical axis), with bubble area encoding production quality. Its 17 points use the same selected recordings and candidate-selection protocols as Tables 1–2.

Prompt aggregation and standardization. For each metric m , let $v_m(s, i)$ denote the score for system s on WSB prompt i . All inputs have complete coverage of the same 192 prompts. Each prompt contributes equally to the system mean:

$$x_m(s) = \frac{1}{192} \sum_{i=1}^{192} v_m(s, i). \quad (10)$$

SongBench Global Avg averages all seven emitted dimensions within a song; SongEval Global Avg averages all five. MuLan and AllMusicCaps use their audio–text similarities, and Q3O uses the prompt-weighted 0–5 score defined in Appendix A.1. We standardize each system-level metric against the same reference set $\mathcal{B}$: all 15 external systems in Tables 1–2, including HeartMuLa and both Suno v6 variants. The two YuE2 settings are excluded from the reference statistics. We compute

$$\mu_m = \frac{1}{15} \sum_{b \in \mathcal{B}} x_m(b), \quad \sigma_m^2 = \frac{1}{15} \sum_{b \in \mathcal{B}} (x_m(b) - \mu_m)^2, \quad z_m(s) = \frac{x_m(s) - \mu_m}{\sigma_m}. \quad (11)$$

This gives each input a common reference mean and variance before weighting. The population standard deviation uses the 15 external system means with divisor 15.

Weighted composites. The quality and alignment composites are

$$Q(s) = \frac{2}{3} z_{\text{SongBench Avg}}(s) + \frac{1}{3} z_{\text{SongEval Avg}}(s), \quad (12)$$

$$A(s) = \frac{1}{3} (z_{\text{MuLan}}(s) + z_{\text{AMCaps}}(s) + z_{\text{Q3O}}(s)). \quad (13)$$

We give SongBench twice the weight of SongEval to emphasize its fine-grained assessment of contemporary song generators. This more recent evaluator covers seven aspects of song quality and uses expert calibration, blind listening, and annotation quality control [77]. Its authors also report stronger agreement with expert Musicality ratings than SongEval [86] on their out-of-distribution test set. SongEval contributes a complementary assessment of song aesthetics. Alignment weights all three standardized inputs equally. The composites summarize standardized metric means under these weights for visualization; the original metrics are the primary reported measurements.

Display coordinates. For display, each composite $C \in \{Q, A\}$ is rescaled to an index on 0–100 axes:

$$I_C(s) = 10 + 80 \frac{C(s) - C_{\min}}{C_{\max} - C_{\min}}. \quad (14)$$

The endpoints are the observed extrema across all 17 settings. The lowest and highest composites map to relative display coordinates of 10 and 90. This positive affine mapping preserves weights, rankings, and pairwise dominance without changing the 15-system standardization reference.

YuE2 and YuE2 (Bo8) obtain quality indices of 82.82 and 88.34, respectively, and alignment indices of 84.59 and 83.89. Both settings exceed each public baseline on both indices. Mureka 9 has the highest quality index (90), while Suno v5 has the highest alignment index (90).

Pareto outlines and bubble areas. A setting t dominates s when $I_Q(t) \geq I_Q(s)$ and $I_A(t) \geq I_A(s)$, with at least one strict inequality. Black outlines identify settings dominated by none of the other 16 settings, using the observed means without rounding. YuE2 (Bo8), Suno v5, and Mureka 9 form the overall empirical Pareto front.

Bubble area separately encodes AudioBox production quality (PQ), with larger areas indicating higher scores; area ratios are not PQ ratios. PQ is excluded from the two-axis Pareto calculation.

Weight sensitivity. We also inspect quality weights from 0.5 to 0.8 for SongBench and replace both global averages with the respective Musicality scores. For either choice of dimensions, YuE2 exceeds every public baseline throughout the 0.6–0.8 weight interval. At the selected 2:1 weighting, the Musicality composite is highest for YuE2 (Bo8), whereas the global-average composite is highest for Mureka 9. The teaser uses global averages to summarize all quality dimensions under the specified 2:1 weighting.

A.5 MUSICALITY AND MODEL SIZE

Figure 11 compares musicality and parameter count for the eight public baselines and both YuE2 settings. Among the two-candidate settings, YuE2 has the highest observed Musicality index (81.63) with 3.58B parameters. It uses 34.0% fewer parameters than HeartMuLa (5.43B, index 78.74), the strongest other public system on this index. Six of the eight baselines have both more parameters and a lower index; DiffRhythm 2 and SongBloom are smaller and have lower indices.

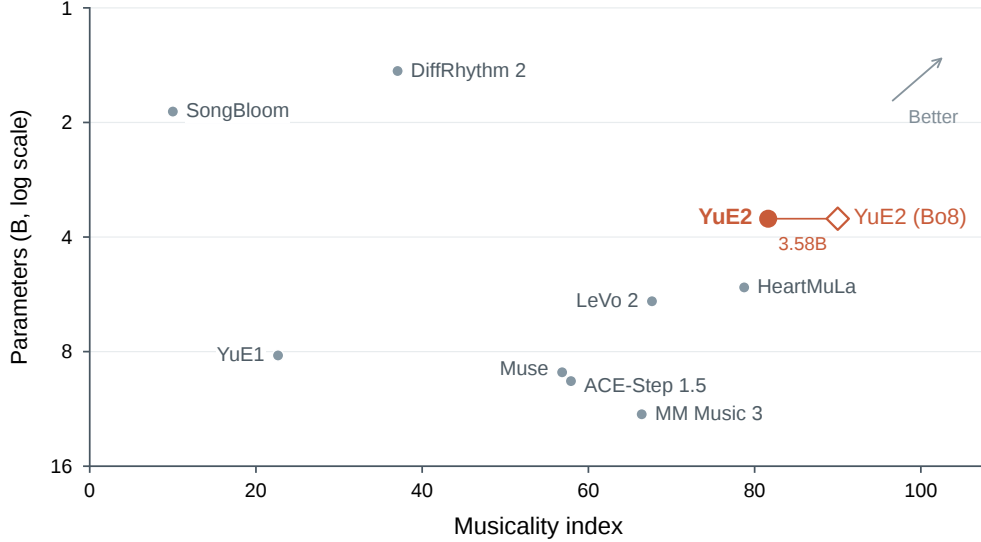


Figure 11: **YuE2 achieves the highest Musicality index among the evaluated public generators, with fewer parameters than six of eight baselines.** Results use 192 WildSongBench prompts. The index combines standardized SongBench and SongEval Musicality scores with 2:1 weights; it is not a percentage. Moving right indicates higher Musicality index; moving up indicates fewer parameters on a reversed logarithmic axis. Circles denote the two-candidate settings; the hollow diamond denotes YuE2 (best-of-8). The connecting segment identifies the same 3.58B-parameter model under these two selection protocols. Counts include generation networks and conditioning encoders, excluding audio tokenizers, VAEs, and vocoders. Appendix A.5 gives the index and counting details.

Musicality index. We combine the Musicality dimensions of SongBench [77] and SongEval [86], using the prompt aggregation and 15-system standardization reference in Appendix A.4:

$$M(s) = \frac{2}{3}z_{\text{SongBench Musicality}}(s) + \frac{1}{3}z_{\text{SongEval Musicality}}(s). \quad (15)$$

We apply the same affine display mapping, with extrema across all 17 WSB settings mapped to 10 and 90; the public subset is not renormalized. The index summarizes observed metric means under these weights, rather than measuring a percentage. Appendix A.4 reports weight sensitivity. HeartMuLa’s raw SongEval score remains included, with its reported evaluator reuse disclosed in Appendix A.3.

Parameter counts. We count the evaluated checkpoints’ language models, generative diffusion networks, and conditioning encoders, including frozen parameters, embeddings, output heads, and retained auxiliary heads. Shared weights count once. We exclude audio tokenizers, VAEs, vocoders, deterministic inverse modules, and fixed buffers; generative diffusion networks within codec pipelines remain included. Counts come from checkpoint tensor shapes and model configurations. MiniMax Music 3’s count covers its released music model and excludes the external caption rewriter, whose size is undisclosed. Best-of-8 uses the same model with a larger candidate budget and a different selection rule (Appendix A.3).

B REPRESENTATION DETAILS

B.1 MERT2 TARGET SYNTHESIS AND TRAINING STAGES

Multi-View Target Synthesis is performed offline before the four-stage MERT2 tokenizer curriculum. The target generator extracts layer 7 of a frozen MuQ encoder [91] and layer 32 of a frozen Qwen2-Audio-Instruct encoder [9], aligns their frame sequences, concatenates the 1,024- and 1,280-dimensional features, and maps them through a learned $2,304 \rightarrow 2,048 \rightarrow 1,024$ fusion network. A four-level residual vector quantizer with 16,384 entries per 1,024-dimensional codebook produces the cached target streams. Separate decoders reconstruct both source representation spaces from the shared quantized state using reconstruction and commitment objectives. This shared target quantizer is distinct from the single-codebook semantic tokenizer used by YuE2.

MERT2 converts 24-kHz mono audio into 128-bin log-mel features and subsamples them to 25 Hz with a ConvNeXt [48] frontend. A 24-layer Conformer [27] uses width 1,024, feed-forward width 4,096, 16 attention heads, and convolution kernel 31. Stage 1, *Foundation Pretraining*, performs masked prediction of the four precomputed synthesized codes. Stage 2-FS, *Full-Song Adaptation*, produces MERT2-FS by retaining bidirectional context and this objective on complete recordings lasting 30–360 seconds. The tokenizer mainline instead uses Stage 2, *Causal Adaptation*; Stage 3, *Supervised Fine-Tuning*, adds lyric connectionist temporal classification (CTC) [25] and mel/chroma objectives; and Stage 4, *Semantic Quantization*, learns the 32,768-entry online clustered vector quantizer (CVQ) adapted from CVQ-VAE [90]. The quantizer uses cosine assignment and online feature anchors to return rarely selected codes to the assignment distribution. During Stage 4, a residual gate ramps the quantized contribution from zero to one over 5,000 steps. The frontend and layers 0–13 remain frozen through step 20,000, after which the full model is unfrozen.

Online codebook updates. Our cosine quantizer adapts CVQ-VAE [90] with $K = 32,768$ entries $e_j \in \mathbb{R}^{32}$. Let p_j be the exponential moving average of code j ’s assignment frequency, with decay $\rho = 0.99$. Each training step samples an anchor a_j from up to 2,048 encoder features gathered across workers, with probability determined by cosine similarity to e_j , then updates

$$e_j \leftarrow (1 - \gamma_j)e_j + \gamma_j a_j, \quad \gamma_j = \exp\left(-\frac{10Kp_j}{1 - \rho} - 10^{-3}\right). \quad (16)$$

Usage is aggregated across workers and the updated codebook is synchronized. The loss combines codebook loss, commitment loss weighted by 0.5, and a contrastive term with temperature 0.07. Codebook utilization is the fraction of entries selected at least once over the validation pass at the selected checkpoint.

Table 9: **MARBLE probe hyperparameters for the reported MERT2 results.** L_k denotes the output of Conformer layer k (numbered 1–24). “All” concatenates the 24 layer representations and applies a learned linear projection to 1,024 dimensions.

Task	MERT2-30s		MERT2-FS	
	Representation	Learning rate	Representation	Learning rate
GTZAN genre	L23	5×10^{-3}	L24	5×10^{-4}
GTZAN beat	L21	1×10^{-3}	L23	1×10^{-3}
GiantSteps key	L4	1×10^{-3}	L23	1×10^{-3}
EmoMusic	All	5×10^{-5}	L24	5×10^{-4}
MTT	L22	1×10^{-3}	L23	1×10^{-3}
MTG Instrument	L14	1×10^{-3}	L12	1×10^{-3}
MTG MoodTheme	L16	1×10^{-3}	L13	1×10^{-3}
MTG Genre	L19	1×10^{-3}	L16	1×10^{-3}
MTG Top50	L13	1×10^{-3}	L22	1×10^{-3}

B.2 SEMANTIC TOKENIZER RETENTION PROTOCOL

Pre-VQ and post-VQ denote the outputs of Conformer layers 12 and 14, respectively, from the same frozen tokenizer (zero-based indexing). Layer 14 is the first Conformer layer after quantization; both representations have 1,024 dimensions at 25 Hz. The 32,768-entry cosine quantizer uses a 32-dimensional codebook, giving a discrete rate of $25 \log_2(32768) = 375$ bits/s. Retention is reported as the ratio of post-VQ to pre-VQ task-probe scores.

We compare this bottleneck with two post-quantization music representations. LeVo 2 [44] applies its generation-time one-codebook residual vector quantization (RVQ) to MusicFM hidden-state index 6 and projects the result back to 1,024 dimensions at 25 Hz. Its 16,384-entry codebook corresponds to 350 bits/s. EnCodec [13] uses the quantizer-decoded 128-dimensional encoder latent from the official 24-kHz model at its 1.5-kbit/s setting: two 1,024-entry codebooks at 75 Hz.

We freeze every encoder and train a separate probe for each representation and task, using seed 1234 and at most 50 epochs. Within each task, data splits, probe architectures, and optimization settings are shared across representations. For MagnaTagATune, we average the selected frame representations over each 30-second excerpt before classification and report macro AUROC and average precision over 50 tags. Emotion reports the mean R^2 across arousal and valence. GTZAN beat tracking uses 10-s excerpts and reports beat F1 with a 70-ms tolerance. For beat tracking we linearly interpolate each tokenizer’s native sequence to the shared 100-Hz label grid—from 25 Hz for the MERT2 tokenizer and LeVo 2 and from 75 Hz for EnCodec. Only the probe’s encoder-facing width changes (1,024 for the MERT2 tokenizer and LeVo 2, and 128 for EnCodec). MTT selects the checkpoint by validation AUROC, and beat tracking by validation beat F1.

B.3 SHEETSAGE2-AR AND SHEETSAGE2-PROBER

SheetSage2-AR builds on audio-to-lead-sheet transcription [16] to jointly predict beat, downbeat, meter, key, chord, musical structure, and lead melody. Melody notes are assigned to vocal (Track 0) or instrumental (Track 1) voices. The benchmarked model is *SheetSage2-AR*; the separate *SheetSage2-Prober* produces its training annotations.

Autoregressive transcriber. SheetSage2-AR pairs a full-context MERT2-FS encoder with a six-layer RoFormer decoder [64]. The encoder backbone is frozen, with trainable low-rank adapters and a learned mixture of its layer outputs. It processes audio in windows of up to 300 seconds. Given encoder output $\mathbf{h}$ and event sequence $\mathbf{e} = (e_1, \dots, e_N)$, the decoder factorizes the conditional distribution as

$$p(\mathbf{e} \mid \mathbf{h}) = \prod_{n=1}^N p(e_n \mid e_{<n}, \mathbf{h}). \quad (17)$$

The benchmark table reports the validation-selected checkpoint. All attributes share one discrete event vocabulary. Events are chronological, with a fixed order when several events have the same onset. The vocabulary is defined as follows.

- **Absolute beat time.** Each beat onset is represented by a 100-Hz-quantized timestamp such as $\langle \text{time_8.48s} \rangle$. Events between beats are located relative to the most recent timestamp.
- **Beat position and meter.** The token $\langle \text{beat_1_of_4} \rangle$ marks a downbeat. Other tokens, such as $\langle \text{beat_4_of_4} \rangle$, give the remaining metrical positions. Supported meters are $2/4$, $3/4$, $4/4$, $3/8$, and $6/8$.
- **Relative subbeat position.** A token such as $\langle \text{subbeat_shift_0.5} \rangle$ advances the event cursor by half a beat. Shifts are quantized to quarter-beat intervals, and no shift is emitted between simultaneous events.
- **Melody pitch and track.** A melody onset such as $\langle \text{pitch_78_track_0} \rangle$ contains a MIDI pitch and source track. The full-melody task includes both tracks; the vocal-melody task includes only Track 0.
- **Note duration.** A pitch may be followed by a quantized token such as $\langle \text{duration_0.5} \rangle$. Durations come from a fixed template; omitting the token denotes the shortest default duration.

- **Chord.** Chords encode root and quality, for example `<chord_F#:maj>`. The vocabulary also includes a no-chord symbol. A chord is emitted when it changes and repeated at downbeats to anchor each measure.
- **Key.** Key changes use 24 root-mode combinations, such as `<key_F#:major>`. The active key is emitted at the first beat and subsequently only when it changes.
- **Musical structure.** Categorical tokens such as `<structure_chorus>` mark the active section at the beginning of a window and at later segment changes.

Table 10 shows the ordering of timestamps, metrical positions, score attributes, melody events, durations, and relative shifts in a short decoded sequence.

Table 10: Example event sequence decoded by SheetSage2-AR.

<code><time_8.48s></code>	<code><beat_4_of_4></code>	<i>// beat</i>
<code><pitch_80_track_0></code>	<code><duration_1.0></code>	<code><subbeat_shift_1.0></code>
<code><time_9.00s></code>	<code><beat_1_of_4></code>	<i>// downbeat</i>
<code><structure_chorus></code>	<code><chord_F#:maj></code>	
<code><pitch_82_track_0></code>	<code><duration_0.5></code>	<code><subbeat_shift_0.5></code>
<code><pitch_73_track_0></code>	<code><duration_0.5></code>	<code><subbeat_shift_0.5></code>
<code><time_9.52s></code>	<code><beat_2_of_4></code>	<i>// beat</i>
<code><pitch_78_track_0></code>	<code><duration_0.5></code>	<code><...></code>

Generating aligned annotations. SheetSage2-AR requires multitask labels with a shared vocabulary and timing convention. Existing music information retrieval (MIR) datasets provide complementary annotations with differing coverage. We first train the non-autoregressive SheetSage2-Prober, fine-tuning MERT2 through low-rank adapters and task-specific prediction heads. Masking losses for unannotated attributes allows training on partially annotated recordings.

Prober supervision includes an expanded lead-sheet corpus based on the data used by SheetSage1 [16], an expanded version of POP909 [75], the training split of HarmonixSet [52], and MIDI-rendered audio from the MIDI AutoLabel Dataset (MALD) [35] and the Segmented Lakh MIDI Subset (SLMS) [17]. MALD supplies automatically inferred chord and key annotations; SLMS supplies section boundaries. We render MIDI with FluidSynth [20] and the FluidR3 GM SoundFont.⁸

The prober predicts frame-level activations, with melody predicted on a subdivided beat grid. Task-specific conditional random field (CRF)-based structured decoding [56] combines these neural scores with temporal constraints to recover discrete labels: rhythm decoding first establishes beats, downbeats, and meter, then key, chord, structure, and melody are decoded on the resulting grid. We apply the prober to a separate corpus of real audio recordings to obtain labels with a common vocabulary and timing convention. SheetSage2-AR is trained exclusively on these generated labels using next-token cross-entropy: 441,094 training recordings (28,434 hours) and 884 validation recordings. A deterministic builder converts the decoded events to ABC, and a parser checks the exported notation.

Comparison with SheetSage2-Prober. Additional experiments compare SheetSage2-AR with SheetSage2-Prober under the same evaluation protocol (Table 11). Although SheetSage2-AR is trained exclusively on labels generated by SheetSage2-Prober, it achieves higher scores on 10 of 15 benchmark–metric pairs. SheetSage2-AR generates the attributes in one chronological event sequence using grammar-constrained autoregressive decoding, replacing the Prober’s task-specific CRF decoders.

Evaluation protocol. Table 8 uses the validation-selected SheetSage2-AR checkpoint. Inference converts audio to mono at 24 kHz and applies grammar-constrained greedy decoding in bfloat16

⁸Official distribution: https://member.keymusician.com/Member/FluidR3_GM/index.html.

Table 11: **SheetSage2-AR compared with SheetSage2-Prober.** The models use the same benchmark cohorts and scoring protocol as Table 8. All scores are percentages; **bold** marks the higher value in each row.

Task	Benchmark	Metric	SheetSage2-Prober	SheetSage2-AR
Beat	GTZAN	F1 $\uparrow$	82.93	86.27
	osu2017	F1 $\uparrow$	92.28	93.01
Downbeat	GTZAN	F1 $\uparrow$	78.74	80.45
	osu2017	F1 $\uparrow$	92.79	92.90
Key	GiantSteps	Score $\uparrow$	78.29	77.73
	GTZAN	Score $\uparrow$	72.62	75.77
Chord	osu2017	Maj/min $\uparrow$	90.43	90.08
	Chords1217	Maj/min $\uparrow$	84.29	83.81
	JAAH	Maj/min $\uparrow$	62.94	64.50
Structure	HarmonixSet	Accuracy $\uparrow$	80.78	80.51
		F1 (0.5 s) $\uparrow$	66.95	67.96
		F1 (3 s) $\uparrow$	81.65	82.86
Melody	RWC-Pop	Vocal F1 $\uparrow$	83.08	82.51
		Full F1 $\uparrow$	75.00	75.29
	Rock Corpus	Vocal F1 $\uparrow$	65.98	67.08

Melody F1 uses a 50-ms onset tolerance and pitch-class matches, with pitches folded across octaves. Prober uses a fixed zero global rhythm offset before downstream processing. Structure F1 measures section boundaries at the indicated tolerance.

(BF16) with batch size one and a maximum output length of 5,120 tokens. Full songs are decoded in 300-second windows with 100-second overlap. Events in the overlap are supplied as a prefix to the next window, and only events in its newly covered tail are retained. We use neither pitch-shift test-time augmentation nor a global timing shift.

For SheetSage2-Prober, we rerun the fixed multitask and beat-conditioned melody models from audio, using 24-kHz mono input, BF16, batch size one, and 300-second multitask windows (30 seconds for GTZAN). Rhythm decoding uses 10-ms resolution and a fixed zero global offset for every dataset. Melody uses 30-second windows with 15-second overlap and four subdivisions per predicted beat, followed by Markov structured decoding, with no subsequent timestamp shift. No reference beats, per-song timing alignment, or pitch-shift augmentation are used. We score the vocal and full-melody outputs separately, before their merger into a lead sheet. Both variants use the same reference tracks, evaluation spans, and metric implementations.

Rhythm results are macro averages of per-track F1 with a 70-ms matching tolerance: 999 GTZAN tracks for beat, 993 for downbeat, and 142 osu2017 [33] tracks for both. Following Beat This! [21], we discard reference and predicted beat/downbeat events before 5 s for every model and use the same reference tracks. We rerun Beat This! under its original inference settings, without DBN postprocessing, and average per-track F1 across its three released training seeds. On the original 993-track GTZAN subset, beat and downbeat F1 match the published means and standard deviations at the reported precision. Our beat evaluation retains the six additional tracks lacking downbeat annotations for every model. Madmom is rescored from saved predictions; the SheetSage1 rhythm entries reuse this backend.

Key evaluation selects the longest predicted key segment and averages the corrected Music Information Retrieval Evaluation eXchange (MIREX) weighted score over 604 GiantSteps and 837 GTZAN tracks.⁹ The scoring assigns 0.5 to same-mode dominant or subdominant errors, 0.3 to relative major/minor errors, and 0.2 to parallel major/minor errors.

Chord results average per-track maj/min chord-symbol recall, weighted by each reference annotation span, over all 142 osu2017 tracks and 1,217 Chords1217 tracks. The osu2017 release fixes

⁹Official MIREX 2025 Audio Key Detection task: https://music-ir.org/mirex/wiki/2025:Audio_Key_Detection.

the evaluated track collection; its beat annotations derive from cleaned game beatmaps, while chord and key annotations were added separately. All available chord predictions use the same references and evaluation time spans. SheetSage1 is independently run on both complete collections; we score its structured harmony output directly. Chords1217 merges the Isophonics, McGill Billboard, and MARL annotations into 1,235 tracks [31]; removing 18 duplicate recordings leaves 1,217. The MARL portion contains 195 USPop and 100 RWC-Pop tracks. The original benchmark defines five cross-validation folds, whereas our SheetSage1, SheetSage2-AR, and SheetSage2-Prober evaluations use fixed checkpoints on the full collection. Madmom has documented training overlap with Chords1217. We rerun the five released ChordFormer [2] models with the official default chord vocabulary and HMM decoder. On osu2017, we average the five models’ component probabilities before decoding; on Chords1217, each track uses only its held-out fold model. Both results are rescored with the same references, time spans, and aggregation as SheetSage2. The implementation extends the chord-structure decomposition code of Jiang et al. [34]. No model is trained or tuned on osu2017. The Chords1217 cross-validation protocol remains distinct from evaluating a single fixed checkpoint.

We additionally evaluate chord transcription on all 113 recordings in JAAH [18]. SheetSage1 uses its released Jukebox-based model and native structured harmony outputs; madmom uses its released CNN feature extractor and CRF chord decoder. Both retain the settings used for the other chord benchmarks. ChordFormer uses the same five-model probability ensemble, default vocabulary, and HMM as on osu2017. All checkpoints and inference settings are fixed for this comparison, without JAAH tuning. The published chord annotations are intersected with each recording’s audio span, without extending shorter annotations or excluding songs. This removes 9.15 seconds of trailing no-chord labels beyond the audio, leaving 24,991.97 seconds of reference annotations. We report `mir_eval` maj/min recall, computed per track and weighted by its complete evaluated reference span. Out-of-vocabulary reference segments are excluded from the per-track denominator; they are not relabeled as no chord. Eligible reference duration, including no-chord labels, is 93.87% for maj/min. Using the untrimmed reference spans changes each reported score by less than 0.03 percentage points.

On JAAH, SheetSage2-AR exceeds ChordFormer by 5.06 percentage points in maj/min recall. The maj/min difference between AR and Prober is 1.56 percentage points.

Structure evaluation uses the 200-song HarmonixSet test split, seven section classes, and a common 0.2-second frame grid. Corpus-level accuracy pools frame counts across all songs; boundary F1 pools matched, reference, and predicted boundary counts at 0.5- and 3-second tolerances. All rescored models use the full reference duration; a non-final `end` label denotes silence.

Melody results average per-song onset/pitch-class F1 over all 100 RWC-Pop songs. Matching uses a 50-ms onset tolerance and 50-cent pitch tolerance after folding pitches across octaves; note offsets are excluded. All systems use the same references and onset criterion, with the fixed Prober timing convention described above.

We additionally evaluate vocal melody on all 200 recordings in Rock Corpus (RS 200) [69], using the released timed vocal-note lists.¹⁰ We retain the supplied note onsets and pitches and use the same 50-ms onset/pitch-class metric, without time alignment or annotation correction. The six recordings whose reference lists contain no pitched notes receive zero F1 and remain in the 200-song macro average. Restricting the average to the 194 nonempty references gives 50.71, 68.02, and 69.15 for SheetSage1, Prober, and AR, respectively. Reference onsets within each measure are derived from metrical positions and bar timestamps. We evaluate onset and pitch-class agreement; note offsets are excluded. SheetSage1 uses its released Jukebox-based model and automatic beats, with six-measure chunks and the existing four-/two-measure fallback. Its single melody output is compared with the vocal reference; AR uses its vocal-melody prompt. Prober uses its vocal output under the fixed inference protocol above. No checkpoints or inference settings are tuned on these annotations.

¹⁰Official melodic transcriptions and timed note lists: https://rockcorpus.midside.com/melodic_transcriptions.html.

C ARCHITECTURE AND TRAINING DETAILS

Training data and annotation. We train MERT2, SheetSage2-AR, and YuE2 primarily on CC0 music and synthetic data. Tokenwave.AI provides most of our synthetic training data under license. Table 12 summarizes the training corpus size for each model.

For YuE2, we adapt the automatic annotation pipeline of YuE [88] with updated annotation models. Qwen3-Omni [80] annotates musical and vocal attributes, including genre, instrumentation, mood, and vocal characteristics. For lyric conditioning, SongFormer [28] supplies section boundaries and functional labels, while Qwen3-ASR [61] transcribes the sung lyrics. These annotations provide the text and lyric conditions y . The corresponding symbolic scores, semantic tokens, and acoustic latents are constructed as described in Section 3.

Table 12: **Training corpus sizes in rounded audio hours.**

Model	Audio hours	Training stage
MERT2	700,000	Foundation Pretraining
SheetSage2-AR	28,400	Autoregressive transcription
YuE2	346,000	Joint symbolic and audio generation

Table 13: **Main YuE2 architecture.**

Property	Value
Parameters	approximately 3.58B
Layers / hidden width / feed-forward width	28 / 2,048 / 6,144
Query heads / key-value groups / head width	16 / 8 / 128
Context length	24,576 positions
Text vocabulary	151,643 entries
Semantic vocabulary	32,768 entries
Acoustic representation	64 dimensions at 25 Hz
Normalization / MLP	RMSNorm [89] / SwiGLU [60]
Training precision	bfloat16

Training tasks. During joint training and annealing, we mix four tasks within the same model (Table 14). The first two retain semantic tokens while varying whether a symbolic score is included, enabling comparison of generation with and without symbolic planning using the same checkpoint. Every task trains acoustic generation with flow matching; next-token loss supervises the included score and semantic tokens.

Table 14: **Four training tasks mixed within one YuE2 model.** The four tasks are sampled in a balanced ratio during joint training and annealing.

Training task	AR prediction targets	Acoustic conditioning
Full generation	Score, then semantic tokens	Text, lyrics, score, and semantic tokens
No symbolic planning	Semantic tokens	Text, lyrics, and semantic tokens
No semantic tokens	Score	Text, lyrics, and score
Direct audio generation	None	Text and lyrics

During training, score and semantic-token sequences derived from the target recording are supplied as conditioning prefixes. Text and lyrics may be dropped separately or together for classifier-free guidance.

Training schedule. We first pretrain an autoregressive model from random initialization and use its 60,000-update checkpoint to initialize the joint AR–NAR model. The NAR Transformer experts are initialized from their AR counterparts. We then perform 40,000 joint-training updates, followed by a 6,000-update annealing stage. We reset the optimizer when starting joint training and retain its state for annealing. Table 15 specifies the learning-rate schedules and context lengths for these three phases.

Table 15: Training phases and optimization schedules for YuE2.

Phase	Updates used	Max. context (positions)	Learning-rate schedule
AR pretraining	60,000	16,384	5,000-step linear warmup to 10^{-4} ; cosine decay with a 10^{-5} floor at step 95,000.
Joint AR–NAR training	40,000	24,576	5,000-step linear warmup to 3×10^{-4} ; cosine decay with a 10^{-4} floor at step 150,000.
Annealing	6,000	24,576	No warmup; cosine decay from approximately 2.726×10^{-4} to 3×10^{-5} over 6,000 updates.

Steps are counted from the start of each phase. The AR and joint-training checkpoints are taken before their scheduled cosine-decay endpoints.

Optimization and packing. All three phases use bfloat16, a global batch size of 256, gradient clipping at 1.0, and 64 H800 GPUs. Adam [37] uses $\beta = (0.9, 0.98)$ and weight decay 0.05 during AR pretraining, and $\beta = (0.9, 0.95)$ and weight decay 0.1 during joint training and annealing. Songs are packed without temporal splitting, and attention and positional indices reset at document boundaries.

Cover supervision. Training uses aligned conditioning inputs and targets from individual recordings, without original–cover pair supervision. Cover evaluation uses the general song-generation checkpoint without cover-specific fine-tuning, on SHS100K works absent from its training corpus (Appendix D.3).

Inference interfaces. Creation generates a score; editing and cover generation supply a score as a fixed prefix.

Table 16: Score conditioning for creation, editing, and cover generation.

Operation	Symbolic score	Generated tokens and latents
Create	$s \sim p(s \mid y)$	$c, z \sim p(c, z \mid y, s)$
Edit	user-modified $\tilde{s}$	$\tilde{c}, \tilde{z} \sim p(c, z \mid y, \tilde{s})$
Cover	$\hat{s}$ transcribed from reference	$c', z' \sim p(c, z \mid y', \hat{s})$

y' combines the source lyrics with the target style description.

D EVALUATION AND STATISTICAL DETAILS

D.1 GENERATED-SCORE AGREEMENT WITH AUDIO

Melody and harmony. For a nonempty reference event sequence R and recovered sequence P , we use

$$S(R, P) = 1 - \frac{D_{\text{edit}}(R, P)}{\max(|R|, |P|)}, \quad (18)$$

where global edit distance assigns cost zero to exact matches and unit cost to insertions, deletions, and substitutions. Both endpoints and every event enter the comparison. The melody sequence uses sounding pitches in semitones: tied notes are merged, separately articulated repeated notes remain separate, and pitches in different octaves are distinct. Each chord is represented by its root and triad quality (major, minor, diminished, augmented, or suspended second/fourth), omitting extensions and inversions. Consecutive chords with the same representation are merged. Unknown recovered labels cannot match known reference labels. These scores compare the resulting note and chord sequences in order, independently of onset times and durations. Edit-distance alignment can resume matching subsequent notes or chords after an insertion or deletion.

Rhythm. We compare consecutive vocal onset intervals, including intervals across rests, in quarter-note beats. The maximum number of one-to-one matches that preserve interval order defines precision and recall over all recovered and target intervals, respectively. A target interval d matches an estimate e when $|e - d| \leq \min(0.125, 0.25d)$ beats. The relative bound prevents doubling short intervals from counting as a match, while the absolute cap limits deviations across long rests. Their harmonic mean is the rhythmic interval F1. This pitch-independent measure compares notated onset spacing on the transcriber’s quantized beat-subdivision grid, without a time warp. Matching rhythmic patterns can receive credit across different melodies. The measure excludes note-off articulation and expressive microtiming.

Global transposition. The primary scores use the specified pitches. For a companion comparison, we shift the reference by one integer number of semitones that maximizes the mean of the available melody and chord similarities. The search includes all shifts that keep reference melody pitches in the MIDI range 0–127; without melody, only a shift modulo 12 is identifiable. Ties prefer the smallest absolute shift, then the smaller signed shift. This single shift applies to melody, chord roots, key, and all sections. Key and form never choose their own shifts. Local octave errors and inconsistent transposition between melody and accompaniment remain errors. The same procedure applies to every score–audio pairing.

Key. We compare the declared score key with the recovered tonic pitch class and major/minor mode, treating enharmonic spellings as equivalent. Alongside exact agreement, we use the key-relation weights used in MARBLE [87]: 1 for the same key, 0.5 for a predicted tonic a perfect fifth above the reference in the same mode, 0.3 for relative major/minor, 0.2 for parallel major/minor, and 0 otherwise. The fifth credit is directional. For a single target key, scores average recovered key regions by their original audio duration, from the first to the last key annotation; internal unknown gaps receive zero. For changing keys, we use the melody correspondence, or chord correspondence if melody is absent, and weight agreement at event onsets by reference event duration. Deleted reference events receive zero. These scores measure agreement with the tonic and mode declared in the score.

Tempo. For each recovered beat interval, let Δq be its length in quarter notes and Δt its duration in seconds. The tempo estimate $\hat{b}$ is the Δq -weighted median of $60\Delta q/\Delta t$, computed before any time alignment. Against target BPM b , we report $|\log(\hat{b}/b)|$ and accuracy under $|\hat{b}/b - 1| \leq 0.08$. A separate accuracy permits $\min_{k \in \{0.5, 1, 2\}} |\hat{b}/(kb) - 1| \leq 0.08$, accounting for half- and double-tempo ambiguity. We report this companion alongside the primary specified-tempo accuracy. Every evaluated score specifies one song-wide tempo.

Musical form. We first merge adjacent same-role fragments, concatenating all their musical events. This tolerates splitting one verse into two annotations while retaining the content of a repeated chorus. For sections i and j , let c_{ij} be the mean of their available melody and chord sequence

similarities. Their pair similarity is $\mathbf{1}[r_i = r_j]c_{ij}$, where r is the section role. Global ordered section alignment assigns cost 1 to an insertion or deletion and cost $1 - \mathbf{1}[r_i = r_j]c_{ij}$ to a substitution. As above, similarity is one minus total cost divided by the longer section count. Unknown roles receive no match credit. Reference sections without observable melody or harmony are unresolved. The score measures section roles and order using their melody and harmony content; boundary timing is evaluated by the separate boundary F1 below. Repeated sustained material with indistinguishable content can match a prolonged occurrence.

Section boundaries. After the same adjacent-role merging, we compare internal section starts, excluding the trivial recording start and end. One-to-one boundary matches within two quarter-note beats define F1 over all target and estimated boundaries. Only a global translation is allowed, estimated as the median target-minus-recovered onset difference of exact pitch matches in the melody sequence alignment; absent anchors give a zero shift. Boundaries do not fit their own alignment, and no scaling or local warp is used. Each control receives the same procedure. Corresponding-audio F1 is 0.8724 with translation and 0.5358 at the original recovered beat origin. The first measures relative section placement, while the second also includes start-position and analyzer beat-origin discrepancies.

Companion results. Allowing one global transposition leaves the corresponding-audio melody and chord scores at 0.9465 and 0.9245. The other same-prompt recording reaches only 0.3152 and 0.4658, respectively (Table 17). The content advantage therefore persists after removing global key and register differences. Individual key or chord scores can decrease slightly because the shared shift optimizes the joint melody/chord objective. Half/double-tempo tolerance raises corresponding-audio tempo accuracy from 0.9869 to 1.0000; the specified-tempo result remains the primary comparison.

Table 17: **Companion comparisons allowing global transposition or tempo ambiguity.** The first five rows share one melody/chord-derived pitch shift per comparison. The last row permits half or double the target tempo, with the same 8% tolerance. Higher is better. Values are prompt means.

Measure	Corresponding	Mismatched	Without score
Melody sequence $\uparrow$	0.9465	0.3152	0.3050
Chord sequence $\uparrow$	0.9245	0.4658	0.4199
Key: exact agreement $\uparrow$	0.9280	0.7527	0.6429
Key: weighted agreement $\uparrow$	0.9489	0.8157	0.7320
Musical form $\uparrow$	0.7798	0.2749	0.2276
Tempo: half/double tolerant $\uparrow$	1.0000	0.7094	0.5654

Denominators and aggregation. All 768 recordings were analyzed without transcription failures. Four of 384 generated scores are truncated and invalid references. Of the remaining 380, one lacks vocal notes and another lacks explicit chords, leaving 379 scores for each melody/harmony metric; rhythm also covers 379. Key and tempo cover 380 scores; form covers 378 because two references have unresolved section boundaries or content. One of these has a zero-duration section marker and is also excluded from boundary F1, leaving 379 boundary targets. Each dimension covers 191 prompts. Missing reference content is undefined; a nonempty reference with empty recovered content scores zero. Unknown predictions remain in the denominator. We average available candidates within each prompt, then weight prompts equally. Paired comparisons retain the same valid cases in both conditions.

Sensitivity to missing reference content. Melody and harmony similarities remain 0.9343 and 0.9122 when all 192 prompts are retained and undefined bounded scores receive zero credit. Under this convention, corresponding-audio means are 0.9414 for weighted key agreement, 0.9766 for tempo accuracy, 0.7696 for form, 0.9277 for rhythm, and 0.8628 for boundary F1. This lower-bound convention does not apply to the unbounded tempo error.

The two-candidate assignment task compares the sum of corresponding scores with the sum under the swapped assignment. Four truncated scores affect three of the 192 WSB prompts, leaving 189

complete pairs. Melody excludes one further prompt lacking vocal notes; chords exclude a different prompt lacking explicit chords. Each task therefore includes 188 prompts, with 187 in common. The correct assignment has a higher similarity sum in all 188 cases for each metric, without ties. This assignment accuracy measures identification of each score’s corresponding recording from the two candidates.

Measurement checks and scope. The metrics were developed on this recording cohort. Music experts checked and corrected each SheetSage2 transcription. Halving both rhythm tolerances leaves corresponding-audio F1 at 0.9396; doubling them gives 0.9506. Boundary windows of one and four beats give F1 0.8709 and 0.8800, respectively. Agreement remains high across the tested rhythm and section-boundary tolerances.

D.2 CONTROLLED SCORE EDITING ON WILDSongBENCH

Setup and aggregation. We use the first generated score for each of the 192 WildSongBench prompts. Key and tempo edits apply to 191 parsable scores; melody, harmony and rhythm edits apply to 190, 181 and 107 scores, respectively. Two seeds per condition yield 3,844 recordings, including unedited controls. All outputs are retained, including 32 at the six-minute cap. Lyrics, style, checkpoint, sampling settings and the main evaluation acoustic decoder remain fixed. We average seeds and edit strengths within each song, then weight songs equally. Rhythm conditions use matched source scores and generation seeds.

Interventions. Key edits transpose all voices, chord roots, bass notes and key declarations by $-5, -2, +2, +5$ semitones. Tempo edits multiply notated BPM by 0.8 or 1.2 and round to the nearest integer; evaluation uses this rounded target. Melody and rhythm edits affect up to four bars of the first chorus. Melody edits raise vocal pitches by two scale steps. Rhythm edits exchange the durations of adjacent quarter and eighth notes with distinct pitches: $(1, 1/2) \leftrightarrow (1/2, 1)$ in quarter-note beats. We select nonoverlapping pairs in score order, preserving pitches, each pair’s start and end, other voices, chords and tempo. The 107 eligible scores contain 210 pairs; eligibility depends only on the source score. Both rhythm conditions use the training ABC serialization rules and retain each source’s unit note length. Independent ABC-to-MIDI checks verify pitch and timing.

Harmony edits process complete chord spans in score order within up to four bars at the start of the first chorus. We shift each root and any specified bass by the first of $(+5, +7, +2, +3)$ semitones that avoids matching either current neighbor in both root and triad quality. Chord quality, timing, and vocal notes remain fixed, and we repeat these substitutions in matching phrases. Bar-aligned windows within sections qualify by identical chord progressions, or by melody pitch-class agreement and onset F1 of at least 0.80 plus chord agreement of at least 0.75 (at least four notes; 1/8-beat onset tolerance). Transferred chords must match the original labels and boundaries, and overlapping substitutions must agree. Edits cover 43.86% of score duration on average.

Attainment and preservation. Harmony adherence and all content-preservation metrics use SheetSage2-AR transcriptions, with BF16 inference, 300-second windows, 200-second overlap and 100-second lookahead. Melody and rhythm adherence, key scores and the MIDI controls use SheetSage2-Prober transcriptions. For editing evaluation, music experts review and correct the transcriptions of generated songs. For melody and harmony attainment and content preservation, up to eight unchanged notes on either side locate local edits, with edited pitches masked during alignment. Melody matching maximizes ordered onset matches within one beat, breaking ties by total onset error; attainment is exact target-pitch accuracy on changed notes. Harmony attainment measures duration-weighted root-and-triad agreement in the designated chorus window, aligned through unchanged vocal notes. Missing events and unresolved contexts score zero for attainment.

For each preservation metric, we exclude undefined scores, average available seeds within each song, and weight evaluable songs equally. For melody and rhythm edits, preservation compares unchanged melody and chord sequences with the source score, penalizing insertions, omissions and substitutions. These comparisons resolve 368 melody-edit and 207 rhythm-edit recordings. Harmony preservation compares the full vocal melody sequence with the source score and measures duration-weighted agreement on unchanged chords outside all edited spans. Chord comparisons use time supported by vocal-note anchors in 170 songs.

Rhythm measurement. We measure onset placement relative to unchanged neighboring onsets, allowing audio to differ in absolute offset and local tempo. We first align full pitch sequences by unit-cost edit distance and identify note correspondences shared by all optimal alignments. Consecutive uniquely matched unchanged onsets define local spans of at most eight quarter-note beats. Within each span, we resolve alignment ties using unchanged onsets within $1/4$ beat of their linearly interpolated source positions, retaining the minimum pitch-edit cost. Edited onsets do not enter this constraint. A correspondence is resolved only when every remaining optimal alignment assigns the same equal-pitch partner without a deletion alternative.

For an edited pair (i, j) , let k be the next resolved unchanged onset after j within eight beats of i , s denote source onsets in quarter-note beats, and t the corresponding audio onsets. The normalized interval is

$$\hat{\Delta}_{ij} = \frac{t_j - t_i}{t_k - t_i} (s_k - s_i). \quad (19)$$

An onset succeeds if this interval is within $1/8$ beat of the edited target and closer to that target than to the original interval. Missing or ambiguous correspondences count as misses in Table 4; coverage is 80.36%. On pairs observable in both conditions (74.95% coverage), mean target-onset error falls from 0.501 to 0.037 quarter-note beats; 96.79% of these pairs move closer to the target.

For each source, we synthesize three instrumental MIDI controls: the target exchange, the unchanged score, and a $1/4$ -beat shift in the opposite direction. The target controls achieve 94.08% onset accuracy; unchanged and opposite-shift controls score zero. Table 18 reports these controls alongside generated songs, with correspondence coverage and conditional accuracy.

Table 18: **Rhythm exchange accuracy and correspondence coverage.** All scores are percentages.

Condition	Onset accuracy	Coverage	Conditional accuracy
Unedited	0.00	88.05	0.00
Edited	73.43	80.36	91.37
MIDI: target exchange	94.08	94.08	100.00
MIDI: unchanged	0.00	94.08	0.00
MIDI: opposite shift	0.00	93.54	0.00

Onset accuracy includes every edited pair; unresolved correspondences score zero. Coverage is the fraction of pairs with resolved correspondences; conditional accuracy uses those pairs. We average pairs within recordings, seeds within sources, and sources equally. All conditions are scored against the edited target. MIDI rows are instrumental controls. Conditional accuracy is defined for 195/214, 188/214, 102/107, 102/107 recordings, respectively.

Key and tempo metrics. Weighted key scores compare recovered tonic and mode with the edited target using the weights in Appendix D.1, averaging over original audio duration between the first and last annotation; unknown intervals score zero.

Tempo uses madmom [6] 0.16.1 recurrent beat activations and dynamic Bayesian network tracking at 100 frames/s, with a 30–300 BPM range and transition parameter 100. We take the median instantaneous BPM between consecutive detected beats; neither the score nor the target enters beat tracking. Acc2 [57] accepts estimates within 4% of the target multiplied by any of $\{1/3, 1/2, 1, 2, 3\}$; Acc1 accepts only the target beat level. Across 764 recordings, Acc2 is 95.68% and Acc1 is 75.92%.

Song quality and lyrics. SongBench [77] quality averages seven dimensions. Greedy audio-only Qwen3-ASR [61] transcripts supply phoneme error rate (PER), using tone-marked pinyin for Chinese and English phonemes for English. Character error rate (CER) retains all scripts and digits after Unicode normalization, case folding and removal of section tags and punctuation. Table 19 reports quality and lyric retention for melody, harmony, rhythm, and key edits.

Table 19: Song quality and lyric retention after editing on WildSongBench.

Condition	Quality $\uparrow$	PER $\downarrow$	CER $\downarrow$
Unedited	6.704	0.184	0.181
Melody	6.717	0.191	0.181
Harmony	6.674	0.179	0.175
Key	6.639	0.190	0.188
Unedited (rhythm)	6.728	0.214	0.197
Rhythm	6.731	0.185	0.175

Quality averages seven SongBench dimensions. PER and CER are phoneme and character error rates. Rhythm rows use matched source scores and generation seeds.

D.3 COVER GENERATION AND VERSION RETRIEVAL

Zero-shot scope and supervision. The evaluated SHS100K [53] works, including their alternate performances, are absent from YuE2’s training corpus. We verified this using CLEWS [59] and Discogs-VINet [4] to check for work-level overlap with the training corpus. YuE2 learns from individual recordings and their aligned annotations, without pairing an original performance with a cover as the target. We use the general song-generation checkpoint without cover-specific fine-tuning. Zero-shot therefore refers to both unseen works and transfer without cover-specific training. The source score is supplied only at inference through the existing conditioning interface; SheetSage2 remains an external analyzer.

The baseline training procedures differ. SongEcho [45] trains a melody encoder and modulation modules on Suno70k while freezing its generative backbone. ACE-Step 1.5 [23] includes multi-task training for audio manipulation, with cover generation supported through quantized source latents. We compare the released systems under their respective training and inference procedures.

Data and selection. SHS100K [53] groups recordings by musical work. We use its public test partition, which lists 10,547 recordings of 1,692 works. The evaluated subset contains 948 source recordings, each from a different work: 947 are labeled English and one Chinese. The retrieval gallery contains 10,546 available recordings from all 1,692 works. Source and style-target candidates are 60–300 s long and come from works with at least three usable versions. Sources must have at least 20 automatically transcribed lyric units, vocal/style annotations, complete symbolic transcriptions, and two annotated alternative versions. Selection also requires a chorus of at least four bars and successful conversion of its score variants. All 948 eligible sources among the 1,349 candidates are included. This cohort consists of vocal recordings with complete automatic transcriptions and two annotated style alternatives.

Source composition and target style. Qwen3-ASR-1.7B [61] transcribes source lyrics and identifies their language; SheetSage2 transcribes melody, chords, beats, key, and form. Qwen3-Omni-30B-A3B-Instruct [80] describes each recording’s genre, instrumentation, voice, tempo, and mood, with song titles, artist names, and lyrics excluded from its instructions. We choose two alternative recordings of the same work to maximize the sum of pairwise Jaccard distances between the source and two target tag sets. Each alternative provides a style description, rather than a waveform to reconstruct. The scores, lyrics, and style descriptions are automatic annotations.

Generation conditions. The full-score condition supplies the source melody-and-chord transcription as a fixed prefix. The condition without chords rebuilds that transcription without the chord timeline, retaining vocal and instrumental melody voices, beats, key, and form; this also changes

chord-dependent serialization boundaries. The condition without a score supplies an empty score prefix but retains lyrics and target style. YuE2 consumes no source waveform during rendering. All three conditions use the same generator checkpoint and 48-kHz stereo decoder used for metric evaluation. Autoregressive sampling uses temperature 1.0, top- p 0.95, top- k 100, and a repetition penalty of 1.2 over 50 tokens. Acoustic generation uses 32 midpoint ODE steps. The acoustic token budget is 25 times the source duration in seconds, capped at 9,000; generation may terminate earlier.

SongEcho [45] uses the source recording to extract its melody condition, with 60 sampling steps, guidance scale 15, and $\omega = 10$. ACE-Step 1.5 [23] uses the released XL supervised-fine-tuned model in native cover mode, with 50 steps, guidance scale 7, shift 3, cover strength 0.8, cover noise strength 0, and language-model thinking disabled. Both receive the source lyrics and target style text. Each method generates two samples for each of the two target styles, for 3,792 outputs per method and 18,960 in total. Retrieval and embedding alignment average all outputs without candidate selection; Q3O coverage is specified below.

Retrieval and metric definitions. Generated audio is the query. The same gallery is used for CLEWS [59] and Discogs-VINet [4], with 10,545 candidates remaining after exclusion of the exact source recording. All remaining recordings of the source work are relevant, including those used to derive target style descriptions. CLEWS uses its released SHS model, segment embeddings extracted with a 5-s hop, and the mean distance of up to ten greedily matched segment pairs, with no segment reused. Discogs-VINet uses its released Discogs-VI model and cosine similarity of normalized track embeddings. Their scores are reported separately.

Average precision averages the precision at every rank occupied by a relevant recording; mAP averages this value across queries. MRR averages the reciprocal rank of the first relevant recording. Hit@ k is the percentage of queries with at least one relevant recording among the first k results, rather than the fraction of all relevant recordings retrieved. Higher values are better for all four measures. The table reports descriptive means over generated queries, with four queries grouped under each source work.

Target-style control and coverage. MuQ-MuLan [91], AllMusicCaps [3], and Q3O [80] compare each generated recording with the target style description used to condition it, excluding lyrics. MuQ-MuLan uses serial full-song inference at 24 kHz; AllMusicCaps averages embeddings from non-overlapping 10-s clips before computing audio-text cosine similarity, as in Appendix A.2. Both metrics cover all 3,792 outputs per method. MuLan reports similarity to the target text. The supplementary AllMusicCaps comparison gives full-score YuE2 a mean of 0.302, compared with 0.349 for SongEcho, the highest value among these five conditions.

Q3O uses the weighted 0–5 score in Appendix A.1, with the target style text as the request. It covers the common available-case subset of 3,286 outputs per method (86.7%), matched by source, target style and seed. These cover 933 works: 710 contribute four outputs per method and 223 contribute two. All active dimensions are valid, and the reported means use the stored weights for these records. The 506 unscored outputs per method are outside this availability-based subset. Qwen3-Omni supplies both target style annotations and Q3O judgments.

Audio quality. AudioBox Aesthetics [70] evaluates production quality (PQ) on the same generated recordings, without candidate selection. SongBench [77] evaluates Musicality on each complete waveform after conversion to 24-kHz mono audio. It covers all 3,792 outputs per method, matched by source, target style, and seed, without candidate selection. Both quality columns report mean scores at three decimal places.

Interpreting the score conditions. Cover generation pairs a fixed source score with a contrasting target style. Our qualitative assessment with music experts links the observed metric ordering to the role of harmony in musical style: retaining the source harmony favors fidelity to the original work, while relaxing it allows greater adaptation to the target style. Retrieval measures work identity; alignment compares the audio with the target description; PQ and Musicality evaluate the audio itself.

After averaging four outputs per work, Musicality is higher than with the full source score for 798 of 948 works (84.2%) when source chords are removed, and for 864 works (91.1%) when no score is supplied or generated. These are descriptive comparisons of work means, with no output selection.

In this fixed-source cover task, the full score yields the strongest work-identity retrieval, while the relaxed conditions achieve higher alignment and quality scores.

D.4 EXPERT LISTENING EVALUATION

Recruitment and coverage. The campaign comprised an online study recruited through X, GitHub, and Hugging Face, and a paid expert panel. The online study contributed 837 retained evaluations from 289 participation sessions, assessing overall quality only. The expert panel comprised 62 annotators with conservatory training or audio AI research backgrounds, including Central Conservatory of Music, Shanghai Conservatory of Music, and Xinghai Conservatory of Music. The Arena export records 351 participations and 4,439 retained pairwise evaluations as of 21 September 2026. The reported Arena preferences use the 3,602 expert evaluations: 2,545 frontier, 636 planning, and 421 architecture evaluations, covering 192 prompts. Each expert evaluation contains separate answers for overall quality, musicality, text alignment, audio quality, vocals, and accompaniment. Within the same listening study, four listeners drawn from the same expert pool also assessed melody and chord progression, as described below.

Evaluation criteria. Table 20 defines the six reported criteria, drawing on the musical, acoustic, and control dimensions in YuE [88] and the song-quality dimensions in SongBench [77]. Overall quality is a direct judgment of the complete song; it is not computed by averaging the other five responses.

Table 20: Definitions of the six expert-listening criteria.

Criterion	Definition
Overall quality	Overall preference for the complete song, considering its composition, performance, sound quality, and fit to the request.
Musicality	Musical appeal and expressive coherence, including memorable melodic phrases, convincing harmony, rhythmic flow, effective arrangement, and development across sections.
Text alignment	Adherence to the supplied style description and lyrics, including the requested genre, mood, instruments, vocal configuration, tempo, and accurate delivery of the words.
Audio quality	Fidelity of the complete recording: audible detail, balanced levels, separation of sound sources, and freedom from unintended noise, clipping, distortion, or synthesis artifacts.
Vocals	Quality of the singing: natural vocal tone, stable pitch and timing, intelligible pronunciation, and expressive phrasing.
Accompaniment	Quality of the instrumental performance: believable timbres, clear articulation, expressive dynamics, and coordination among instruments and with the voice.

Listening and consistency checks. System identities were hidden and A/B presentation order was randomized. The expert protocol required at least 30 seconds of listening to each song. Every batch of 32 questions contained five hidden consistency checks, with at least four correct responses required to pass the batch. The checks used either identical audio on both sides, which required a tie, or a previously presented pair with the audio order reversed, which required the same preference after restoring the model identities. The analysis retains non-check expert responses that were neither rejected nor part of an invalidated group at the export cutoff. Online responses are retained only from accepted rounds, excluding check items and responses marked for exclusion from voting.

Generation and selection. The planning comparison holds the checkpoint, prompt, candidate budget, and decoder fixed. The LM+DiT architecture baseline follows Qwen-Music [81] and uses the same training-data volume as the unified MoT. Its LM and DiT each have 1.7B parameters, for

3.4B in total compared with MoT’s 3.58B. The LM and MoT’s AR component use the same token budget, batch size, and number of training updates; the DiT is trained for 900,000 updates. Both systems use the same semantic tokenizer, acoustic VAE, and symbolic planning method. The main architecture comparison uses melody-and-chord symbolic planning, the same 192 prompts, and two candidates per prompt. Each candidate is transcribed four times with ASR, and its lowest PER is retained. The lower-PER candidate is then selected for each prompt. YuE2 (best-of-8) selects from eight candidates, prioritizing SongBench Musicality, then Q3O, then PER.

D.4.1 COMPLETE FRONTIER PREFERENCES

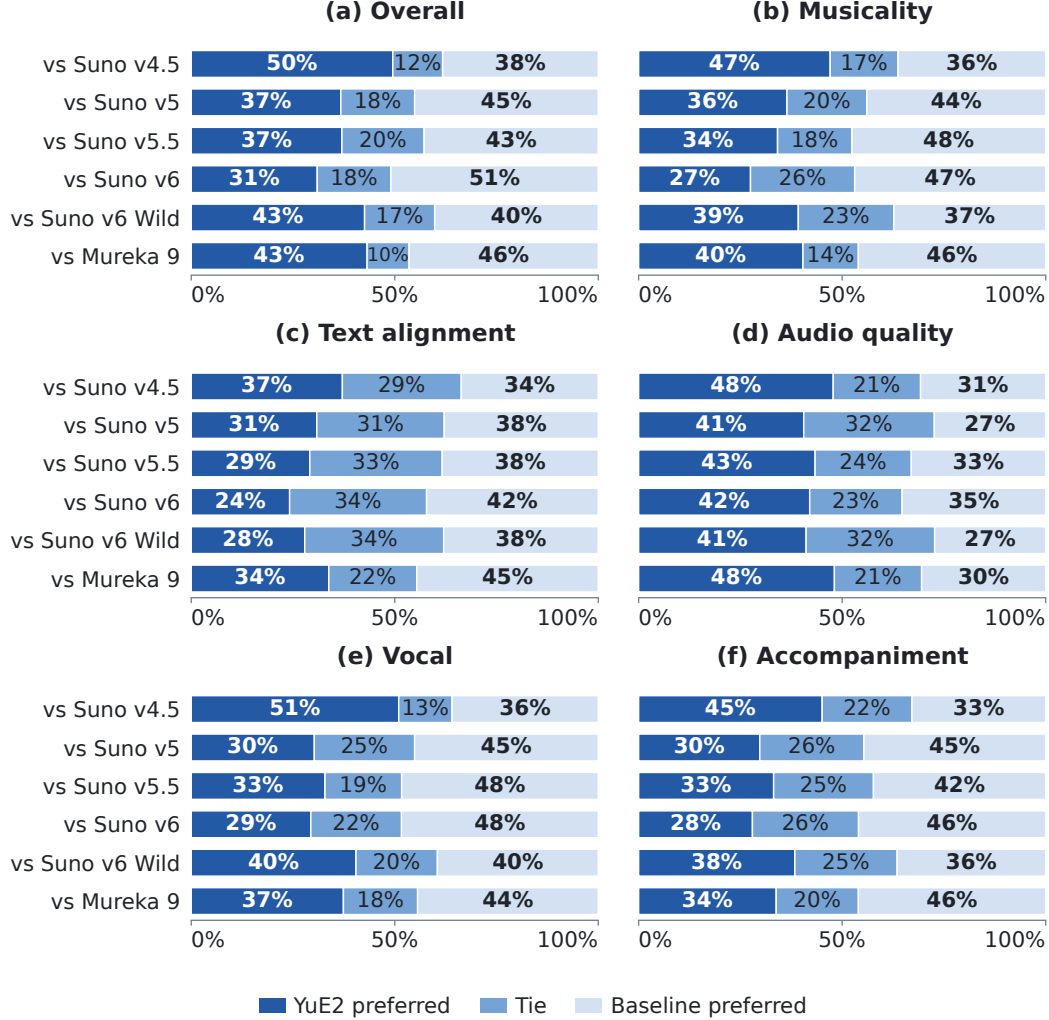


Figure 12: **Expert preferences for YuE2 versus six proprietary song generators across all six criteria.** Audio-quality preferences favor YuE2 against all six baselines; overall preferences favor it in two comparisons. YuE2 uses melody-and-chord planning and selects the candidate with fewer lyric errors from two generated songs. Baselines are Suno v4.5 [68], v5 [65], v5.5 [62], v6 and v6 Wild [66], and Mureka 9 [51]. Dark blue favors YuE2, light blue favors the named baseline, and medium blue denotes ties. Baseline order is shared by the left and right panels. Each bar sums to 100% of judged responses, excluding unable-to-judge answers. Counts and CR1 intervals are in Appendix D.4.4.

For audio quality, experts prefer YuE2 to each of the six proprietary baselines more often than they prefer the reverse (Figure 12). Overall preferences favor YuE2 in two of the six comparisons. Preferences for musicality, text alignment, vocals, and accompaniment vary by baseline.

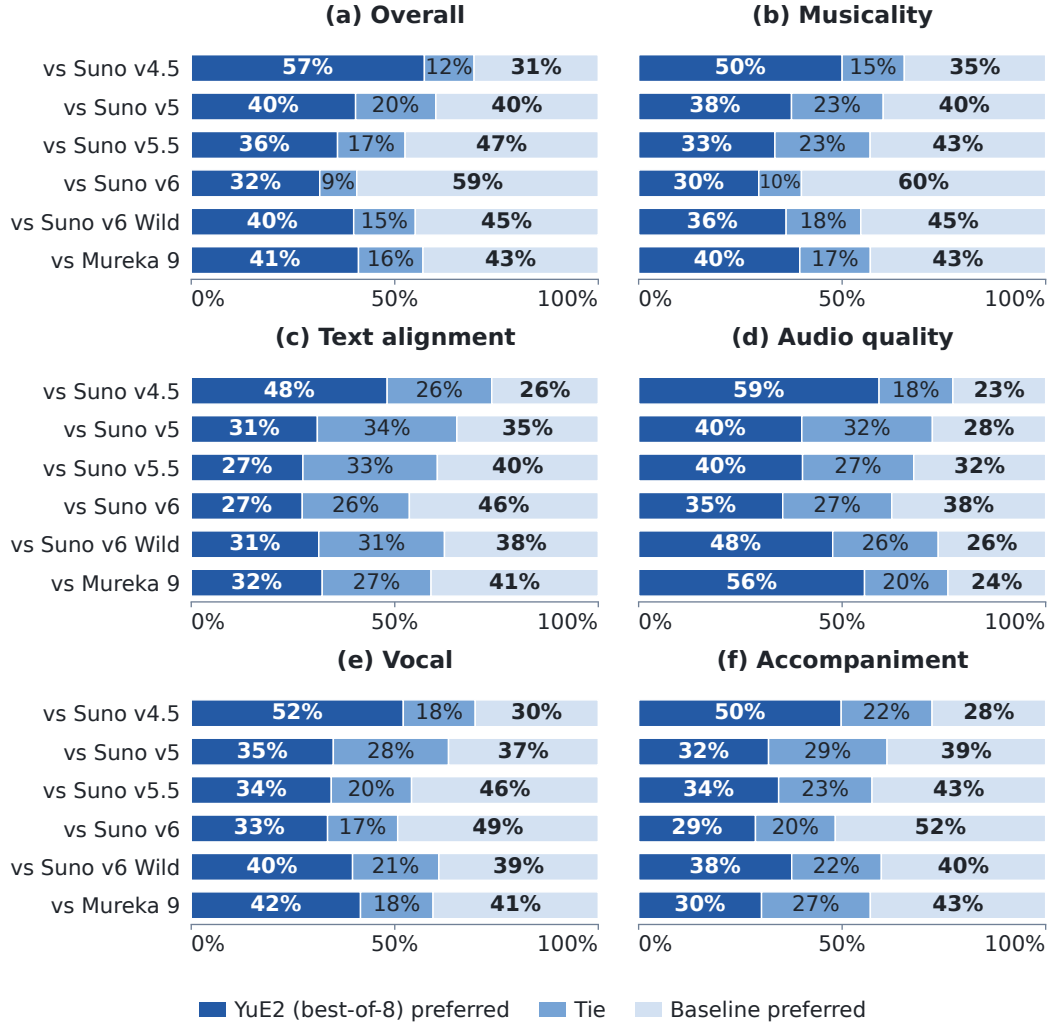


Figure 13: **Expert preferences for YuE2 (best-of-8) versus six proprietary song generators across all six criteria.** Overall preferences favor best-of-8 over Suno v4.5, are nearly balanced against v5, and favor v6 over best-of-8. Audio-quality preferences favor best-of-8 in five of six comparisons. YuE2 uses melody-and-chord planning and selects from eight candidates by musicality, prompt adherence, then lyric accuracy. Baselines are Suno v4.5 [68], v5 [65], v5.5 [62], v6 and v6 Wild [66], and Mureka 9 [51]. Dark blue favors YuE2 (best-of-8), light blue favors the named baseline, and medium blue denotes ties. Baseline order is shared by the left and right panels. Each bar sums to 100% of judged responses, excluding unable-to-judge answers. Counts and CR1 intervals are in Appendix D.4.4.

Best-of-8 receives more overall preferences than Suno v4.5 and nearly balanced preferences against Suno v5. Suno v6 receives more overall preferences (Figure 13). For audio quality, experts prefer best-of-8 in five of the six comparisons; Suno v6 receives more preferences in the remaining comparison. The six-baseline average audio-quality preference is 58.9% (Appendix D.4.4).

D.4.2 COMPLETE PLANNING AND ARCHITECTURE COMPARISONS

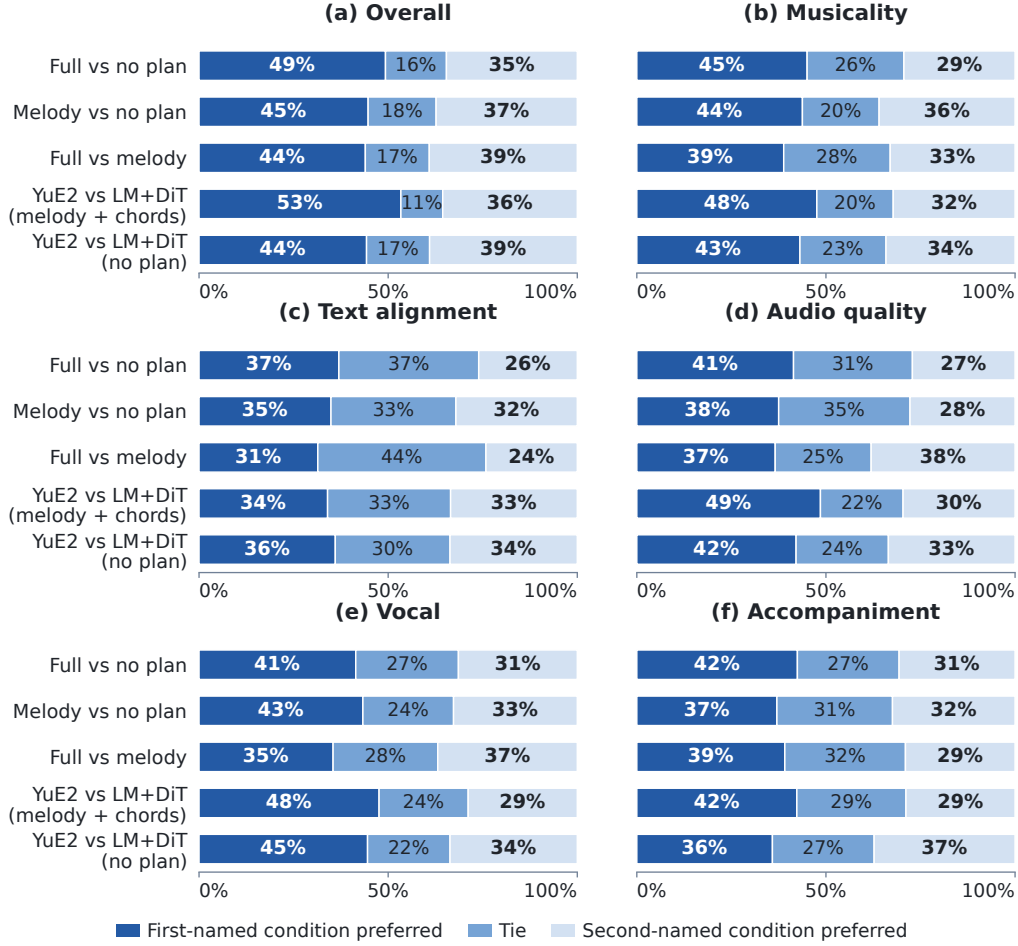


Figure 14: **Expert preferences across six criteria for symbolic-planning ablations and YuE2 versus separate LM+DiT.** Full planning receives more preferences than no planning on every criterion; unified generation is preferred overall both with and without planning. Each panel contains five direct comparisons. Full planning includes melody and chords; melody-only planning omits chords. The first three rows compare planning conditions within YuE2. The last two compare YuE2’s AR–NAR Mixture-of-Transformers [14] with a separate language model and diffusion Transformer (LM+DiT), with and without planning. Dark blue favors the first-named condition in each row; light blue favors the second. Medium blue denotes ties. Each bar sums to 100% of judged responses, excluding unable-to-judge answers. Counts and CR1 intervals are in Appendix D.4.4.

Melody-and-chord planning receives more preferences than generation without planning in all six dimensions, with the largest preference margins in overall quality and musicality (Figure 14). Melody-only planning also receives more preferences than generation without planning across the six dimensions. Adding chords to the melody plan increases overall, musicality, text-alignment, and accompaniment preferences; audio quality and vocals are nearly balanced.

Preference shares favor the unified MoT over separate LM+DiT for overall quality, musicality, audio quality, and vocals both with and without symbolic planning. With melody-and-chord planning, accompaniment preferences also favor the MoT, while text alignment is nearly balanced. Without planning, text-alignment preferences also favor the MoT, while accompaniment is nearly balanced. The statistical tables distinguish these direct comparisons from averages over the two planning conditions or the two architecture conditions.

D.4.3 MELODY AND CHORD EVALUATION

Four listeners drawn from the same expert pool evaluated 100 song pairs generated with and without symbolic planning, holding the prompts, checkpoint, candidate budget, and decoder fixed. Each listener assessed 50 pairs, and each pair received two evaluations from different listeners. Presentation order was randomized and system identities were hidden. Listeners assessed melody, chord progression, and overall quality separately. Melody concerns phrase shape, development, and memorability; chord progression concerns harmonic coherence, movement, and fit with the melody. Overall quality has the meaning given in Table 20. Figure 8 reports the melody and chord-progression judgments. Each criterion has 200 judgments, and preference shares use the corresponding criterion’s denominator.

D.4.4 STATISTICAL PROTOCOL

We recover model identities from the recorded A/B presentation order. For each criterion, W and L count preferences for the first- and second-named systems, and T counts ties. Figure percentages use $n = W + T + L$ as the denominator; the tables separately report unable-to-judge answers as U . The tie-adjusted preference is

$$S = \frac{W + 0.5T}{W + T + L}. \quad (20)$$

Average comparisons weight model pairs equally and responses equally within each pair. The frontier averages cover the six proprietary baselines. The planning average combines melody-and-chord versus no planning with melody-only versus no planning. The architecture average combines the planning and no-planning comparisons.

Expert standard errors use two-way CR1 clustering by evaluator and song; online standard errors use one-way CR1 clustering by anonymous participation session. Pointwise 95% t intervals use the smaller evaluator/song cluster count minus one as the degrees of freedom for experts, and the session count minus one for online responses. Two-sided tests compare S with 0.5. We report each test’s unadjusted p -value. Best-of-8 average audio-quality preference is 58.9% (CR1 95% CI 55.6–62.2%; $p = 1.70 \times 10^{-6}$). Tables 21 and 22 provide the counts, CR1 intervals, and p -values for every expert direct and average comparison.

Table 21: **Expert pairwise preferences.** W , T , and L count preferences for the first-named system, ties, and preferences for the second-named system. $n = W + T + L$; U counts unable-to-judge answers. Preference and intervals are percentages. Intervals and tests use two-way CR1 clustering by evaluator and song. p tests a tie-adjusted preference of 50% (two-sided).

Criterion	Counts					Preference (%)		p
	W	T	L	n	U	Estimate	CR1 95% CI	
YuE2 (best-of-8) vs Mureka 9								
Overall	83	32	87	202	0	49.0	39.7–58.3	0.8316
Musicality	80	35	87	202	0	48.3	40.7–55.8	0.6477
Text alignment	65	54	83	202	0	45.5	37.8–53.3	0.2523
Audio quality	111	41	48	200	2	65.8	57.8–73.7	2.26×10^{-4}
Vocal	84	36	82	202	0	50.5	43.1–57.9	0.8930
Accompaniment	61	54	87	202	0	43.6	35.0–52.1	0.1369
YuE2 (best-of-8) vs Suno v4.5								
Overall	122	26	65	213	3	63.4	57.7–69.1	2.26×10^{-5}
Musicality	108	33	75	216	0	57.6	50.9–64.4	0.0269
Text alignment	103	55	56	214	2	61.0	55.4–66.6	2.76×10^{-4}
Audio quality	127	39	49	215	1	68.1	62.8–73.5	1.65×10^{-8}
Vocal	112	38	65	215	1	60.9	54.1–67.8	0.0024
Accompaniment	107	48	60	215	1	60.9	55.0–66.8	5.02×10^{-4}

Continued on next page

Table 21 continued

Criterion	Counts					Preference (%)		<i>p</i>
	<i>W</i>	<i>T</i>	<i>L</i>	<i>n</i>	<i>U</i>	Estimate	CR1 95% CI	
YuE2 (best-of-8) vs Suno v5								
Overall	86	42	85	213	0	50.2	44.0–56.5	0.9402
Musicality	80	48	85	213	0	48.8	43.2–54.4	0.6765
Text alignment	66	73	74	213	0	48.1	42.3–54.0	0.5212
Audio quality	85	68	59	212	1	56.1	50.3–62.0	0.0400
Vocal	74	60	78	212	1	49.1	41.8–56.3	0.7956
Accompaniment	68	62	83	213	0	46.5	40.8–52.1	0.2159
YuE2 (best-of-8) vs Suno v5.5								
Overall	78	36	103	217	1	44.2	38.4–50.0	0.0518
Musicality	73	51	94	218	0	45.2	39.0–51.3	0.1220
Text alignment	59	71	85	215	3	44.0	39.0–48.9	0.0184
Audio quality	87	59	70	216	2	53.9	48.9–59.0	0.1221
Vocal	75	43	100	218	0	44.3	38.9–49.6	0.0359
Accompaniment	75	50	93	218	0	45.9	40.8–51.0	0.1095
YuE2 (best-of-8) vs Suno v6								
Overall	66	19	124	209	1	36.1	28.0–44.2	0.0012
Musicality	62	22	126	210	0	34.8	27.5–42.1	1.06×10^{-4}
Text alignment	57	55	97	209	1	40.4	33.9–46.9	0.0047
Audio quality	74	56	79	209	1	48.8	40.9–56.7	0.7631
Vocal	70	36	103	209	1	42.1	34.4–49.8	0.0444
Accompaniment	60	41	108	209	1	38.5	30.9–46.1	0.0038
YuE2 (best-of-8) vs Suno v6 Wild								
Overall	87	33	98	218	1	47.5	40.2–54.7	0.4883
Musicality	79	40	99	218	1	45.4	39.3–51.6	0.1406
Text alignment	68	67	82	217	2	46.8	40.9–52.7	0.2759
Audio quality	103	56	57	216	3	60.6	55.7–65.6	7.51×10^{-5}
Vocal	86	46	85	217	2	50.2	44.1–56.4	0.9404
Accompaniment	82	48	88	218	1	48.6	42.7–54.6	0.6455
YuE2 vs Mureka 9								
Overall	92	22	99	213	1	48.4	40.8–55.9	0.6655
Musicality	86	29	98	213	1	47.2	38.4–55.9	0.5219
Text alignment	72	46	95	213	1	44.6	38.2–51.0	0.0977
Audio quality	101	45	64	210	4	58.8	50.3–67.3	0.0422
Vocal	80	39	95	214	0	46.5	37.6–55.3	0.4306
Accompaniment	72	43	98	213	1	43.9	37.0–50.8	0.0836
YuE2 vs Suno v4.5								
Overall	101	25	78	204	3	55.6	49.2–62.0	0.0831
Musicality	96	34	74	204	3	55.4	49.4–61.4	0.0782
Text alignment	75	59	68	202	5	51.7	46.4–57.0	0.5137
Audio quality	98	44	63	205	2	58.5	52.9–64.2	0.0039
Vocal	105	27	74	206	1	57.5	50.6–64.4	0.0327
Accompaniment	92	45	67	204	3	56.1	49.7–62.5	0.0607

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Table 21 continued

Criterion	Counts					Preference (%)		<i>p</i>
	<i>W</i>	<i>T</i>	<i>L</i>	<i>n</i>	<i>U</i>	Estimate	CR1 95% CI	
YuE2 vs Suno v5								
Overall	79	39	97	215	1	45.8	38.1–53.6	0.2825
Musicality	78	42	94	214	2	46.3	40.2–52.3	0.2196
Text alignment	66	67	81	214	2	46.5	40.5–52.5	0.2496
Audio quality	86	68	58	212	4	56.6	49.8–63.4	0.0583
Vocal	65	53	97	215	1	42.6	36.2–48.9	0.0226
Accompaniment	64	55	96	215	1	42.6	35.8–49.3	0.0323
YuE2 vs Suno v5.5								
Overall	77	42	89	208	0	47.1	39.9–54.4	0.4295
Musicality	71	38	99	208	0	43.3	35.9–50.6	0.0729
Text alignment	60	67	79	206	2	45.4	39.0–51.8	0.1529
Audio quality	88	48	67	203	5	55.2	47.8–62.5	0.1647
Vocal	68	39	100	207	1	42.3	35.6–49.0	0.0244
Accompaniment	69	51	88	208	0	45.4	37.0–53.9	0.2825
YuE2 vs Suno v6								
Overall	65	38	107	210	1	40.0	32.2–47.8	0.0131
Musicality	58	54	99	211	0	40.3	33.6–46.9	0.0050
Text alignment	51	71	89	211	0	41.0	35.5–46.4	0.0017
Audio quality	87	47	73	207	4	53.4	43.2–63.5	0.5060
Vocal	62	47	102	211	0	40.5	32.9–48.1	0.0158
Accompaniment	59	55	97	211	0	41.0	33.6–48.4	0.0186
YuE2 vs Suno v6 Wild								
Overall	89	36	84	209	2	51.2	44.0–58.4	0.7416
Musicality	82	49	78	209	2	51.0	45.7–56.3	0.7185
Text alignment	58	71	79	208	3	45.0	39.0–50.9	0.0949
Audio quality	86	66	57	209	2	56.9	49.0–64.9	0.0852
Vocal	85	42	83	210	1	50.5	43.4–57.5	0.8929
Accompaniment	81	53	77	211	0	50.9	43.7–58.2	0.7933
YuE2 vs YuE2 with melody-only planning								
Overall	91	35	81	207	0	52.4	46.9–57.9	0.3819
Musicality	80	58	68	206	1	52.9	47.3–58.5	0.3033
Text alignment	65	92	50	207	0	53.6	48.2–59.0	0.1845
Audio quality	75	52	78	205	2	49.3	42.8–55.8	0.8219
Vocal	73	57	76	206	1	49.3	44.2–54.4	0.7762
Accompaniment	81	66	60	207	0	55.1	50.1–60.0	0.0443
YuE2 vs YuE2 without planning								
Overall	104	34	73	211	1	57.3	52.1–62.6	0.0070
Musicality	95	54	62	211	1	57.8	51.6–64.0	0.0141
Text alignment	78	78	55	211	1	55.5	49.5–61.4	0.0732
Audio quality	87	66	57	210	2	57.1	51.0–63.3	0.0229
Vocal	87	57	66	210	2	55.0	49.1–60.9	0.0927
Accompaniment	90	57	65	212	0	55.9	50.4–61.4	0.0357

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Table 21 continued

Criterion	Counts					Preference (%)		<i>p</i>
	<i>W</i>	<i>T</i>	<i>L</i>	<i>n</i>	<i>U</i>	Estimate	CR1 95% CI	
YuE2 with melody-only planning vs YuE2 without planning								
Overall	97	39	81	217	0	53.7	47.9–59.5	0.2064
Musicality	95	44	78	217	0	53.9	47.4–60.4	0.2316
Text alignment	75	71	69	215	2	51.4	46.1–56.7	0.5973
Audio quality	81	75	60	216	1	54.9	49.9–59.8	0.0532
Vocal	94	52	71	217	0	55.3	50.6–60.0	0.0295
Accompaniment	80	66	70	216	1	52.3	46.7–57.9	0.4110
YuE2 vs Separate LM+DiT								
Overall	111	23	74	208	1	58.9	52.4–65.4	0.0084
Musicality	99	42	67	208	1	57.7	52.3–63.1	0.0065
Text alignment	70	67	69	206	3	50.2	45.6–54.9	0.9162
Audio quality	100	45	61	206	3	59.5	54.6–64.4	3.06×10^{-4}
Vocal	99	49	60	208	1	59.4	55.3–63.4	2.69×10^{-5}
Accompaniment	88	60	60	208	1	56.7	51.7–61.8	0.0097
YuE2 without planning vs Separate LM+DiT without planning								
Overall	93	35	82	210	2	52.6	47.1–58.1	0.3456
Musicality	91	48	72	211	1	54.5	49.8–59.2	0.0602
Text alignment	76	64	71	211	1	51.2	45.2–57.2	0.6950
Audio quality	88	51	70	209	3	54.3	49.5–59.2	0.0807
Vocal	94	46	71	211	1	55.5	49.2–61.7	0.0882
Accompaniment	76	57	79	212	0	49.3	42.2–56.4	0.8416

Table 22: **Average expert preferences.** Model pairs receive equal weight. Preference and intervals are percentages; n counts judgments and U counts unable-to-judge answers. Intervals and tests use two-way CR1 clustering by evaluator and song. p tests a tie-adjusted preference of 50% (two-sided).

Criterion	Counts		Preference (%)		p
	n	U	Estimate	CR1 95% CI	
YuE2: average over six proprietary systems					
Overall	1259	8	48.0	43.8–52.2	0.3498
Musicality	1259	8	47.2	43.6–50.8	0.1263
Text alignment	1254	13	45.7	42.4–49.0	0.0110
Audio quality	1246	21	56.6	51.6–61.6	0.0105
Vocal	1263	4	46.6	42.7–50.6	0.0971
Accompaniment	1262	5	46.7	42.8–50.6	0.0920
YuE2 (best-of-8): average over six proprietary systems					
Overall	1272	6	48.4	44.8–52.0	0.3813
Musicality	1277	1	46.7	43.4–50.0	0.0484
Text alignment	1270	8	47.6	44.7–50.6	0.1111
Audio quality	1268	10	58.9	55.6–62.2	1.70×10^{-6}
Vocal	1273	5	49.5	45.8–53.2	0.7931
Accompaniment	1275	3	47.3	44.4–50.3	0.0730
Architecture: average of planning and no-planning comparisons					
Overall	418	3	55.8	51.8–59.7	0.0050
Musicality	419	2	56.1	52.2–60.0	0.0027
Text alignment	417	4	50.7	47.1–54.3	0.6928
Audio quality	415	6	56.9	53.2–60.5	3.98×10^{-4}
Vocal	419	2	57.4	53.5–61.3	3.86×10^{-4}
Accompaniment	420	1	53.0	49.0–57.0	0.1353
Planning: average of melody-and-chord and melody-only comparisons					
Overall	428	1	55.5	51.1–59.9	0.0153
Musicality	428	1	55.9	51.5–60.2	0.0086
Text alignment	426	3	53.4	49.2–57.7	0.1143
Audio quality	426	3	56.0	51.9–60.1	0.0048
Vocal	427	2	55.1	51.3–59.0	0.0098
Accompaniment	428	1	54.1	49.6–58.6	0.0752

E FULL SCORES FOR THE PLANNING COMPARISON

The following scores give the full-song context for the excerpts in Figure 9. Both songs use the same prompt and lyrics. The score for the planned song starts from its generated plan; for the song without planning, SheetSage2 transcribed the audio. After audio generation, music experts checked and corrected both scores against the recordings for display. These corrections were not used to regenerate audio.

Figure 15 displays a full musical score with symbolic planning. The score is written for a song in G major (one sharp) and 4/4 time, with a tempo of 70. The score is divided into sections: Intro (measures 1-8), Verse (measures 9-25), Chorus (measures 26-32), and Outro (measures 33-50). The upper staff shows the vocal melody, and the lower staff shows the instrumental accompaniment. Chords are indicated above the staves. Colored highlights mark specific features: purple for repeated lyric openings (e.g., measures 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50), orange for instrumental responses (e.g., measures 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50), blue for rhythmic passages (e.g., measures 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50), and green for the closing line (measures 49, 50).

Figure 15: **Full score with symbolic planning.** The upper and lower staves show vocal and instrumental melodies. Purple marks repeated lyric openings; orange, blue, and green mark instrumental responses, rhythmic passages, and the closing line discussed below.

Figure 16: **Transcription of the song generated without symbolic planning.** Colors follow Figure 15; gray additionally marks the harmonic transition discussed in Section 4.2.

Instrumental responses, rhythm, and ending. In the planned song, the **orange** fills connect vocal phrases and prepare a smoother chorus entry; comparable responses are sparse in the unplanned song after the introduction. In the planned song’s **blue** passages, syncopation and varied note lengths animate the vocal phrasing, while the unplanned song’s continuous sixteenth-note texture sounds more monotonous. The **green** closing line is reprised in the planned song, allowing the ending to settle; the unplanned song’s single statement feels abrupt.